

Affording Not to Farm: Remittances after a Large Natural Disaster*

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We study how remittance responses to a large natural disaster affect farm production, the value of crop and livestock output. We combine pre- and post-earthquake household surveys from Nepal in a triple-difference instrumental-variables design that interacts spatial differences in earthquake intensity with exchange-rate-induced shifts in the value of remittances. We find that in high-intensity earthquake areas, higher remittance inflows crowd out farm production in the short run, while consumption remains stable. Higher inflows also make households less likely to farm at all. The production decline is concentrated in grains and meat, and the adjustment runs through capital inputs, chiefly livestock inputs, handling and upkeep, and seed, together with the value of households' own agricultural labor. Remittances do not appear to crowd out public aid. Non-remitting households instead lean on it in the absence of transfers from abroad. From a policy perspective, remittances appear to ease short-run budget constraints enough that recipient households can afford not to farm, scaling back production while smoothing consumption and without drawing on additional public support. They are therefore not a substitute for policies aimed at sustaining farm production after a large shock.

Keywords: Remittances, Natural Disasters, Farm Production, Agriculture, Nepal

JEL-Classification: F24, O15, Q12, Q54, D13

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1 – Introduction

In many low- and middle-income countries, money sent home by migrants is what households live on: it finances consumption, absorbs risk, and funds what they produce. Recorded inflows to these economies reached US\$651 billion in 2022 (Ratha et al. 2024). A large shock tests that role hardest, cutting into a household’s usual income just when the claims on that income are greatest. Nepal is a particularly useful setting in which to ask what the remittance response to such a shock does to farm production, because it is highly dependent on both remittances and agriculture: 22% of households have at least one member living abroad, while agriculture, broadly defined, remains the main production sector for 64% of the working-age population (National Statistics Office 2023).¹ Remittances are also among Nepal’s largest external resource flows, accounting for more than 20% of GDP in recent years (Appendix Figure A1). Remittances and farm production are therefore jointly central to household welfare in Nepal.

Much of the existing literature studies how migration and remittances shape origin-household welfare, investment, and labor allocation in normal times (Mobarak, Sharif, and Shrestha 2023; Yang 2008b; Mishra, Kondratjeva, and Shively 2022). More recent work also shows that exchange-rate-induced changes in migrant income opportunities can have persistent effects on origin-area development (Khanna et al. 2026). Much less is known about how remittance inflows affect production choices when households are suddenly hit by a major shock. That question matters more as extreme events become more frequent and as climate change, demographic pressure, and migration reshape local economies (McAuliffe and Triandafyllidou 2021; Guha-Sapir, Hoyois, and Below 2017; IPCC 2014). The question is sharpest at the moment of the disaster itself, when a household faces urgent and competing demands on cash, labor, and assets at once and transfers from abroad are one of the resources it can draw on. Whether that liquidity sustains farm production or substitutes for it is an open question, with direct implications for how disaster relief and recovery programs are designed.

This paper studies how the remittance response to a large disaster changes farm production

1. These figures are from the 2021 National Population and Housing Census, the most recent comprehensive source; the 64 percent is the census category covering agriculture, forestry and fishing. Patterns were similar around our estimation window: at the 2011 census, 25.4% of households reported at least one member absent abroad (Central Bureau of Statistics 2011a), and 55.8% of households received remittances in 2010/11 (Central Bureau of Statistics 2011b).

choices. We focus on the 2015 Nepal earthquake, which caused more than 9,000 deaths and displaced more than 500,000 households (Government of Nepal, National Planning Commission 2015). The earthquake created an immediate need for liquidity in affected areas. Extra remittances could, in principle, help households maintain production by financing agricultural inputs and labor. But they could equally ease short-run pressure on the budget and make it feasible to scale back farming, an option households without remittance support do not have. The distinction matters because production and welfare need not move together: a household can smooth its consumption with remittance income while scaling back the farm activity that income was, in part, supposed to protect.

We therefore ask three questions. Does the remittance response to a large disaster preserve farm production, or displace it? If production falls, which margins absorb the adjustment: what households eat, what they spend on inputs, or the labor they supply to their own land? And is that adjustment driven by the transfers themselves, or by differential access to public support? These questions are empirical, and answering them requires variation in the remittance response that is not itself a consequence of how a household was affected.

Identifying that effect is difficult because remittances are not randomly assigned. Migration opportunities, migration risks, and the returns to migration differ across households, so remittance receipt is closely linked to underlying household choices and constraints (Mobarak, Sharif, and Shrestha 2023; Theoharides 2018; Shrestha 2019). Remittances can also respond endogenously to adverse household shocks and distress, precisely because they often serve an insurance role (Yang and Choi 2007; Yang 2008a).

We address this with a triple-difference instrumental-variables design that combines two sources of variation. The first is spatial: wards differ in earthquake intensity, measured by Modified Mercalli Intensity, so the shock itself splits otherwise comparable localities. The second is monetary: households send migrants to different countries, and the currencies of those countries moved differently against the Nepalese rupee over the period, so the rupee value of a given transfer varies across households for reasons they do not choose. Interacting the two with the post-earthquake period isolates how the remittance response to the earthquake, rather than remittance receipt as such, changes production. Every remittance term is instrumented by its exchange-rate counterpart, following Yang (2008b).

The empirical analysis draws on two bodies of evidence. The primary analysis rests on the household surveys, which measure remittances, production, and the input and consumption margins for the same households, paired with the exchange-rate variation just described. Supporting evidence comes from a second, non-monetary source: annual remote-sensing land cover for every ward in Nepal, which covers the whole country but records only land in use, combined with predetermined migration exposure built from the 2001 and 2011 censuses.

We find that higher remittance inflows in more severely affected areas reduce farm production in the short run, and the magnitude is substantial: for households already producing and receiving, a ten percent rise in remittances corresponds to roughly a four percent fall in output. Nineteen percent of household-years report no production at all, so this estimate aggregates an extensive and an intensive margin whose relative weight is not pinned down by the data (Chen and Roth 2024). Separating them shows the response is concentrated at the extensive margin: higher inflows reduce the probability of farming at all rather than the scale of output among those who continue. Across components the decline is concentrated in grains and meat, the two categories that depend most on continuing outlays for seed, feed, and animals; the rest are negative but cannot be distinguished from zero. Consumption is unchanged: the aggregate and all seven of its food groups are small and statistically indistinguishable from zero. Households adjust instead through farm inputs and labor, cutting back the cash-intensive inputs that sustain output, chiefly livestock inputs, handling and upkeep, and seed, along with the value of their own agricultural labor, while spending on hired labor is indistinguishable from its pre-earthquake level. Public assistance is the one margin that does not respond to remittances: earthquake aid rises sharply in high-intensity areas, and higher inflows do not reduce it — so remittances crowd out farm production without crowding out public support. The mediation exercise points the same way, tying the production effect more closely to capital and labor than to consumption or public assistance.

Taken together, these results describe a pattern of post-disaster adjustment in which remittances relax the need to maintain farm production rather than financing its recovery. Remittance income may also shift labor supply within the household once there is an outside source of cash, consistent with evidence from Nepal on left-behind members (Lokshin and Glinskaya 2009; Phadera 2016). [Section 6](#) develops this interpretation, asks how much of the production decline the observed margins account for, and sets out what the pattern implies for how recovery is

supported.

We contribute foremost to the literature on remittances and shocks. Remittances often rise when origin households are hit by adverse events and can stabilize external inflows after disasters (Yang and Choi 2007; Yang 2008a; Eldemerdash and Landis 2023), but much less is known about what those additional transfers do to farm production. We provide causal evidence on that margin and show that larger remittance responses go with lower production, not stronger recovery: remittances relax the need to maintain agricultural activity rather than finance a return to it.

Second, we speak to work on agricultural production in developing countries, where relaxing credit and risk constraints raises input use, technology adoption and output (Karlan et al. 2014; Duflo, Kremer, and Robinson 2008). Our results mark a different margin. After a severe disaster, liquidity from abroad need not raise production; it coincides with lower input use, so remittances operate differently from interventions aimed directly at farm investment.

A third contribution is to separate two outcomes that are often read together: smoothing consumption does not imply preserving production. Remittances are known to insure recipient households against income shocks and reduce consumption instability (Gubert 2002; Yang and Choi 2007; Combes and Ebeke 2011; Ebeke and Combes 2013). Households everywhere want to smooth consumption and draw on whatever instruments are available, borrowing, drawing down assets, and adjusting what they produce; our results indicate that remittances are one of those instruments after a large shock. Post-disaster remittance responses can therefore be consistent with consumption smoothing even as farm production shrinks.

A fourth contribution is about public assistance, which remittances also do not displace. Households meet disasters with whatever is available, from insurance and credit (Gallagher and Hartley 2017) to buffer assets such as livestock (Fafchamps, Udry, and Czukas 1998), and governments extend their own support after such events (Klomp and Hoogezand 2018). Less is known about whether transfers from abroad displace that support, which bears directly on how relief is targeted. We observe both margins for the same households and find they operate side by side: earthquake aid rises sharply in high-intensity areas, and higher inflows do not reduce what a household receives. Differential access to public support therefore cannot account for why recipients produced less.

Finally, we provide evidence on mechanisms. The reduced-form decomposition is the primary

evidence, estimating each margin separately under the same assumptions as the headline estimate; a structured mediation exercise following Imai, Keele, and Tingley (2010) supports it. The two agree, tying the production decline much more closely to lower capital and labor use than to lower consumption.

Section 2 describes the Nepali context, Section 3 the data, Section 4 the empirical strategy and identification, Section 5 the results and mechanisms, Section 6 what they imply for how recovery is supported, and Section 7 concludes.

2 — Background

2.1. Migration, Remittances, and Agriculture

Two features of the Nepali economy matter for what follows: where its migrants go, and how much of household income their transfers represent. Migration to India has long been important and requires no permit, but over recent decades migration to the Gulf, Malaysia, Japan, and South Korea has expanded substantially (Seddon, Adhikari, and Gurung 2002; World Bank 2011; Shrestha 2019). That expansion is what gives the design its variation: the destinations differ in currency, and the Nepalese rupee is pegged to the Indian rupee but floats against the others, so an Indian migrant and a Malaysian migrant transmit markedly different exchange-rate movements to households that are otherwise alike. Remittances have meanwhile become a leading source of household liquidity and of foreign exchange (Sapkota 2013); Appendix Figure A1 plots the aggregate series.

2.2. The 2015 Nepal Earthquake

The April 2015 earthquake caused extensive human and economic losses and left many households facing urgent spending needs and severe disruption (USGS 2017; Goda et al. 2015). Two features of it matter for the design. It was unanticipated, so neither migration decisions nor destination choices already reflected it, and its intensity varied sharply over short distances, so neighbouring wards with similar economies were shaken differently. Recent work on the same event documents large and persistent pressure on local economic and institutional outcomes (Pathak and Schündeln 2022, 2026).

3 — Data and Variables

3.1. Household Surveys

We combine the third round of the Nepal Living Standards Survey (NLSS, 2010/11) with the Household Risk and Vulnerability Survey (HRVS, 2016–2018) (Central Bureau of Statistics 2011b; World Bank 2020). The NLSS is an independent sample and provides our pre-earthquake baseline, while the HRVS is a three-year post-earthquake panel that resurveys the same households in 2016, 2017 and 2018. Both are multi-topic household surveys covering the household roster, migration and remittances, agricultural production, consumption expenditure and household finance. Both also identify the ward where the household lives, so we can attach earthquake intensity to each household. Because the NLSS sample is independent, no household appears both before and after the earthquake, and we therefore compare households within a location rather than within a household.

The unit of observation is the household-year. We build the panel from the household rosters and the migration modules of both surveys, and we link each migrant back to their roster record wherever the survey reports a member identifier. Production, consumption and the input blocks are all household quantities, so the household-year is the level at which the outcome actually varies. Remittances belong to the individual migrant, and we aggregate them to the household by summing over its migrants. The exchange rate is aggregated the same way, as the simple average of the rates faced by the household’s migrants, which is the migrant-count weighting used by Yang (2008b). A household with migrants in two countries therefore enters once, carrying the average of the two rates.

Our estimation sample contains 22,354 household-years, drawn from 11,460 distinct households in 875 wards, 712 VDCs and 71 districts. The 2011 baseline contributes 5,899 household-years, and the post-earthquake rounds contribute 5,515, 5,425 and 5,515, so the panel is close to balanced across the four rounds. Of the 22,354 household-years, 11,280 have at least one migrant and 7,271 have a migrant abroad; the latter are the household-years that carry exchange-rate variation.

3.2. Variables and Descriptive Statistics

Our main outcome is farm production, the value of a household's crop and livestock output over the survey year, measured over the seven Food Consumption Score food groups the survey records on the production side (WFP 2024). Three further outcome blocks are built the same way. Consumption is the food a household purchases, over the same seven groups, so the two aggregates are defined over identical foods. Capital inputs cover the cash-intensive agricultural inputs, labor inputs cover hired agricultural labor together with the imputed value of the household's own, and public assistance covers government and program transfers including earthquake relief. All monetary variables are converted to 2018 Nepalese rupees.

The design combines three measured objects: the transfer, the shock, and the instrument that moves the first without responding to the second. Remittances are what the household received from its migrants over the survey year, summed across them and zero for households with no migrant; because a substantial share receive nothing, the benchmark enters remittances as $\log(1 + R)$. Earthquake intensity is the Modified Mercalli Intensity at the household's ward, taken from the USGS ShakeMap for the 25 April 2015 event, and a ward is high-intensity when its MMI exceeds the sample median. [Appendix Figure A3](#) maps that variation against district-average remittances and production, and shows that intensity varies within districts as well as across them, which is the variation the design uses. The exchange rate is the rate between the Nepalese rupee and the currency of the migrant's destination, averaged over the year before the household's interview, so it varies across households through where they send migrants and over time through currency movements. [Appendix Table A2](#) reports the destinations and how much each rate moves, and [Appendix Table A5](#) the estimates under two alternative rules for households with migrants in more than one country.

About forty percent of household-years receive remittances, and [Appendix Table A1](#) compares those that do with those that do not. Receiving households are smaller and far more likely to be headed by a woman, the expected composition effect of adult male out-migration, and the two groups face the same average earthquake intensity, so the remittance split does not stand in for exposure to the shock. They are statistically indistinguishable in production, consumption and capital inputs, and differ on two margins: receiving households supply fewer household labor

hours and receive less public assistance. Neither gap is causal, since the two groups differ on many margins at once, but they locate where the groups differ before any structure is imposed. The sample is smallholder throughout, with median annual production of NPR 31,000.

Appendix A.1 gives the full variable definitions and the survey modules they are built from, the coding conventions behind the household exchange rate, the descriptive comparison in full, and the distribution of migrants across destination countries.

4 – Empirical Strategy

4.1. Specification

We estimate a triple difference: earthquake intensity, interacted with the post-earthquake period, interacted with exposure to migration income. Exposure is measured two ways, which trade coverage against content. Predetermined exposure is a shift-share built from the censuses, which observe every ward in every year but carry only land in use and no transfers. Actual receipts come from the surveys, which cover a sample of wards but measure production, inputs, labor, consumption, public assistance, and remittances themselves. We begin with the censuses.

Predetermined exposure weights each ward’s 2001 destination shares across ten migrant-destination groups by destination-specific migration growth between the 2001 and 2011 census waves (Central Bureau of Statistics 2001, 2011a), computed outside the ward’s own district, following Khanna et al. (2026) and Yang (2008b).² We pair this with the National Land Cover Monitoring System of Nepal (NLCMS 2022), which classifies Landsat imagery for the whole country each year and which we aggregate to cropland hectares per ward, giving an annual panel of every ward from 2000 onward.

C_{wt} denotes log cropland area measured in hectares in ward w in year t , HI_w indicates above-median earthquake intensity measured by Modified Mercalli Intensity, and HSS_w indicates above-median shift-share exposure. We estimate one coefficient per year, so that the response can

2. Appendix C gives the construction. Under the 75-district structure in place at the time, each ward’s shifter comes from migration growth in the other 74 districts, which prevents a ward’s own migration trajectory from determining its predicted exposure.

be dated rather than assumed:

$$C_{wt} = \alpha_w + \tau_{d(w)t} + \sum_{\substack{t=2001 \\ t \neq 2014}}^{2018} \gamma_t (\mathbf{1}(t) \times HI_w) + \sum_{\substack{t=2001 \\ t \neq 2014}}^{2018} \beta_t (\mathbf{1}(t) \times HI_w \times HSS_w) + \Phi_{wt} + \varepsilon_{wt}, \quad (1)$$

where α_w are ward fixed effects and $\tau_{d(w)t}$ district-year fixed effects. Bold denotes fixed effects throughout the paper. The displayed coefficients are those the tables report; everything else that enters estimation is collected into a single symbol, here

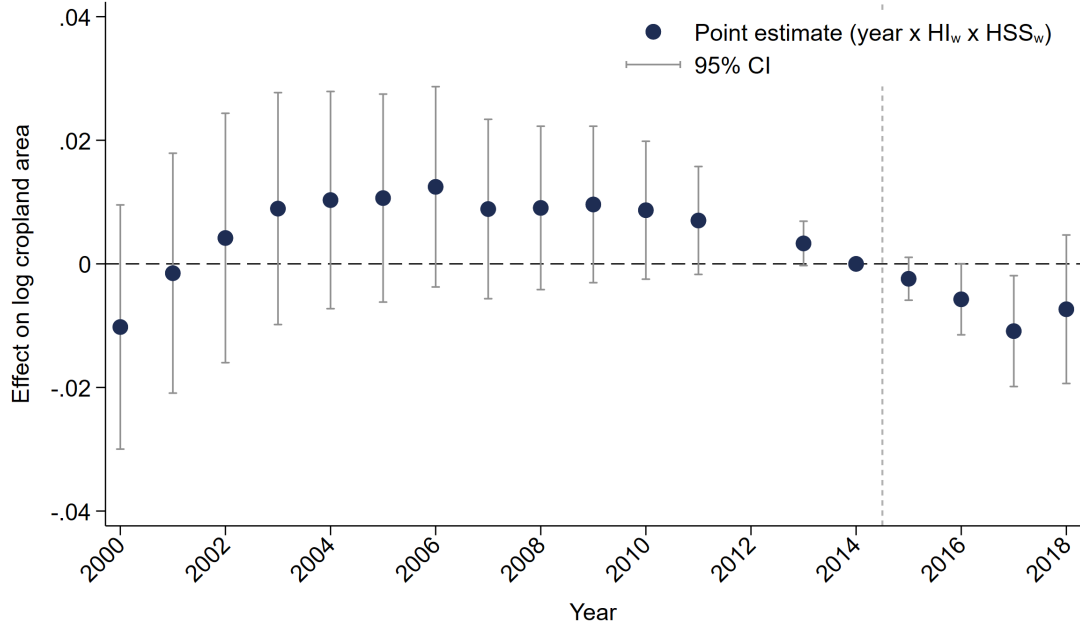
$$\Phi_{wt} = \sum_{\substack{t=2001 \\ t \neq 2014}}^{2018} \pi_t (\mathbf{1}(t) \times HSS_w) + \sum_{\substack{t=2001 \\ t \neq 2014}}^{2018} \mathbf{1}(t) (W'_w \kappa_t + S_w \mu_t),$$

which holds three things: the year-specific lower-order interaction of HSS_w , without which each β_t cannot be read as a triple difference; the ward socioeconomic controls W_w from the 2011 census, covering ward households, population, the poverty index and the agricultural household share, each interacted with year; and $S_w = \sum_c z_{wc,2001}$, the sum of the ward's baseline destination shares, also interacted with year. Conditioning on the share sum is standard in shift-share designs with incomplete shares (Borusyak, Hull, and Jaravel 2022). Because S_w is the ward's overall migration rate rather than unity, variation in HSS_w would otherwise conflate the level of exposure with its destination composition; holding S_w fixed leaves identification resting on composition among wards with comparable overall exposure. The main effects of HI_w and HSS_w are absorbed by α_w .

The estimation sample spans 2001 to 2018. We exclude 2012, for which a Landsat 7 sensor failure left parts of the country unscanned, and use 2014, the last year before the earthquake, as the omitted year. Each β_t is therefore identified from the change in cropland between 2014 and year t in high-intensity, high-exposure wards, relative to the contemporaneous change elsewhere. The coefficients estimated for $t < 2015$ provide a test of the identifying assumption, while those for $t \geq 2015$ trace the timing and persistence of the response. We also estimate a pooled specification that replaces the year indicators with a single 2015–18 indicator, reported in Panel A of Table 1. Standard errors are clustered at the district level throughout.

4.2. Supporting Evidence from Land Cover

Figure 1 – Cropland event study by earthquake intensity and migration shift-share



Notes: The solid dots represent point estimates of the interaction $\text{year} \times HI_w \times HSS_w$; the light lines show 95 percent confidence intervals; and the light vertical bar marks the 2015 earthquake. HI_w indicates above-median ward earthquake intensity. HSS_w indicates above-median shift-share migration exposure. The omitted year is 2014, the year immediately before the earthquake. The dependent variable is log ward cropland area, constructed as $\log(C + 0.09)$, where C is cropland hectares from the NLCMS data (NLCMS 2022). A negative point estimate means that cropland area is relatively lower in high-migration-shift-share wards in high-earthquake-intensity locations in that year. The NLCMS data do not report 2012 because several pixels were not scanned due to the failure of a Landsat 7 sensor. The specification includes ward fixed effects, district-year fixed effects, year-interacted core socioeconomic controls, and year interactions with the sum of baseline destination shares. Standard errors are clustered at the district level.

Figure 1 plots the β_t from (1). The pre-earthquake coefficients are indistinguishable from zero, so the two groups of wards were not already diverging. From 2015 onward they are negative in every year, and they become more negative over time. Table 1 reports the same specification under alternative transformations of cropland area: Panel B lists the year-specific coefficients behind the figure, and Panel A the pooled coefficient, which is negative and significant throughout. Appendix C.1 enters exposure continuously rather than as an indicator, in Appendix Figure C1 and Appendix Table C1, and Appendix C.2 repeats the whole exercise at the VDC, the geography the household regressions absorb, in Appendix Figure C2 and Appendix Table C2.

Two things follow. Cropland in earthquake-affected wards fell after 2015, with no differential

Table 1 – Cropland area and migration shift-share exposure after the earthquake

	Dependent variable: alternative transformations of cropland area C						
	$\log(C + 0.09)$	$\text{lhs}(C)$	C (ha)	$\log(C)$ for $C > 0$	calibrated $\log(C)$		
					$x=0.1$	$x=0.5$	$x=1$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Panel A. Pooled post-earthquake years 2015–18</i>							
2015–18 x HI_w	0.007 (0.006)	0.006 (0.006)	1.152* (0.681)	0.008 (0.006)	0.006 (0.006)	0.006 (0.006)	0.006 (0.006)
2015–18 x HI_w x HSS_w	-0.007* (0.004)	-0.007** (0.003)	-0.411 (0.265)	-0.008** (0.003)	-0.007** (0.003)	-0.007** (0.003)	-0.007** (0.003)
<i>Panel B. Separate post-earthquake years 2015–18</i>							
2015 x HI_w x HSS_w	-0.002 (0.002)	-0.003* (0.001)	-0.158 (0.105)	-0.004** (0.002)	-0.002 (0.002)	-0.002 (0.002)	-0.002 (0.002)
2016 x HI_w x HSS_w	-0.006* (0.003)	-0.006** (0.003)	-0.413* (0.213)	-0.007*** (0.003)	-0.006** (0.003)	-0.006** (0.003)	-0.006** (0.003)
2017 x HI_w x HSS_w	-0.011** (0.005)	-0.010** (0.004)	-0.650** (0.322)	-0.011** (0.004)	-0.009** (0.004)	-0.009** (0.004)	-0.009** (0.004)
2018 x HI_w x HSS_w	-0.007 (0.006)	-0.009* (0.006)	-0.421 (0.475)	-0.012* (0.006)	-0.012* (0.006)	-0.011* (0.006)	-0.010* (0.006)
N	617922	617922	617922	611598	617922	617922	617922
District	75	75	75	75	75	75	75
Ward and district-year FE	✓	✓	✓	✓	✓	✓	✓
Core controls	✓	✓	✓	✓	✓	✓	✓

Notes: Dependent variables are ward-level NLCMS cropland area, transformed as indicated; C denotes cropland hectares. HI_w indicates above-median ward MMI and HSS_w indicates above-median predicted migration shift-share exposure. The shift-share combines pre-earthquake destination shares with leave-one-district-out destination growth across Nepal's 75 districts. Both panels use the 2001–18 sample and omit 2014. Panel A pools the 2015–18 triple interaction; Panel B displays the separate 2015–18 triple interactions. All specifications include ward fixed effects, district-year fixed effects, year-interacted core socioeconomic controls, and year interactions with baseline destination-share totals. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors, reported in parentheses, are clustered at the district level across 75 districts.

trend before it, and the decline was larger where predetermined migration exposure was higher. Exposure here is not a migration rate that could respond to local post-earthquake conditions: it is built from shares fixed in 2001 and growth computed outside the ward's own district, so neither the timing nor the gradient is driven by migration decisions taken after the shock.

What this design cannot do is measure the object of interest. Cropland records land in use, not output, and says nothing about the inputs, labor, consumption or transfers behind a change in it, and the censuses record no remittances at all. A residual identification concern remains besides: the baseline shares are a ward choice, and places that sent migrants to the fastest-growing

destinations may differ in labor-market access or infrastructure in ways that shaped agricultural adjustment independently of remittances. Leaving the ward’s own district out of the growth factors addresses the shock side of that problem but not the share side (Goldsmith-Pinkham, Sorkin, and Swift 2020; Borusyak, Hull, and Jaravel 2022, 2025). We need data that measure both remittances and production, and a design that does not rely on the shares.

4.3. Identification and First Stage

The household surveys supply what the censuses lack: production and its components, consumption, capital, labor and public assistance, the remittances each household received from each migrant, and where each migrant went, which is what makes a household-specific exchange rate available as an instrument. We therefore move from the ward-year to the household-year, and from predetermined exposure to the transfer itself. The two settings differ in one respect that matters: a shift-share built from 2001 shares is predetermined, whereas the amount a household receives after the earthquake is a choice. More productive households may find it easier to finance a migrant, distress migration compensates for low income or scarce land, and transfers respond to the same shocks that affect production. Selection therefore runs in both directions and the sign of the bias is not known in advance, which is why we instrument.

4.3.1. The instrumental variable

y_{hwt} denotes farm production for household h , which lives in ward w , in year t , and R_{hwt} the remittances that household receives, summed over its migrants; a household with no migrant carries a remittance of zero. Each ward lies in a village development committee, written $v(w)$. Here we display the pooled specification rather than the year-specific one, because it is the form the instrumented estimate takes and the form the first stage is built on:

$$y_{hwt} = \alpha_{v(w)} + \tau_t + \gamma(2016-18 \times HI_w) + \beta(2016-18 \times HI_w \times R_{hwt}) + \Psi_{hwt} + \varepsilon_{hwt}, \quad (2)$$

$\alpha_{v(w)}$ are village development committee (VDC) fixed effects and τ_t year fixed effects.³ Ψ_{hwt} plays the role Φ_{wt} plays in (1), collecting everything estimated but not reported: the three lower-order remittance terms R_{hwt} , $HI_w \times R_{hwt}$, and $2016-18 \times R_{hwt}$, together with the household controls X_{hwt} and ward controls Z_w described below. Both displayed coefficients are reported. γ is the difference-in-differences effect of high earthquake intensity after the shock, evaluated at zero remittances, and so describes households without remittance income; β measures how that effect changes with remittance inflows.

We impose the fixed effects at the VDC, a choice bounded on both sides. A ward effect is not feasible, because the rounds are repeated cross-sections drawn from different survey instruments and few wards appear both before and after the earthquake.⁴ A district effect would absorb the identifying contrast, since intensity is measured at the ward and varies within districts, as [Appendix Figure A3](#) shows. The VDC is therefore the finest grouping that retains a usable pre-post comparison. [Appendix Table A9](#) re-estimates the benchmark totals with district fixed effects; the results are substantially unchanged in direction and significance, with production, capital and labor all declining and consumption and public assistance not. [Appendix C.2](#) rebuilds the land-cover evidence at the VDC, so both parts of the paper can be read at that geography.

Controls enter at two levels and are reported as separate rows in the tables. Household controls X_{hwt} cover the ethnicity, age group and marital status of the head. We exclude household size, because a migrant's departure mechanically reduces the size of the household left behind, so conditioning on it would absorb part of the channel we are measuring. Ward controls Z_w are fixed at their 2011 census values and cover ward area, poverty, population and the agricultural shares of months and occupations. [Appendix A.1](#) lists both sets in full.

We also estimate a year-specific version of (2), replacing the 2016–18 indicator with separate indicators for 2016, 2017, and 2018 and taking 2011 as the base, with Ψ_{hwt} expanded to match. This is (1) applied to the household design, and it serves the same purpose: it shows whether the effect

3. Nepal's administrative hierarchy over this period runs district $\rightarrow$ VDC $\rightarrow$ ward, with 75 districts. Despite the name, a VDC is a geographic administrative unit rather than a body, and is roughly comparable to a municipality in other countries. The data identify the ward, but we absorb the VDC because only units observed both before and after the earthquake contribute to the comparison, and far more VDCs than wards are observed in both periods.

4. Only 11 of the 875 wards we observe are present in both periods, against 69 of 712 VDCs. Because most sampled VDCs contribute a single ward, the VDC effect is close to a ward effect in practice, and ward characteristics enter as controls in any case.

appears immediately and whether it persists. The production tables report those coefficients in Panel B; for the mechanism outcomes the main text reports the pooled estimate and [Appendix A.5.3](#) the year-specific one.

Both versions are estimated by two-stage least squares, following Yang (2008b). Every remittance term is treated as endogenous and instrumented by its exchange-rate counterpart: R_{hwt} by ER_{hwt} , $HI_w \times R_{hwt}$ by $HI_w \times ER_{hwt}$, $2016-18 \times R_{hwt}$ by $2016-18 \times ER_{hwt}$, and the term of interest $2016-18 \times HI_w \times R_{hwt}$ by $2016-18 \times HI_w \times ER_{hwt}$. The fitted values from these first stages, written $\hat{R}_{hwt}$ in the tables, replace R_{hwt} throughout (2). Instrumenting only the triple-difference term would leave the lower-order remittance terms endogenous, so all four are instrumented together. Exchange-rate movements shift the local-currency value of remittances for migrant households while being plausibly orthogonal to production choices in rural Nepal, conditional on fixed effects and controls. Standard errors are clustered by district throughout.

4.3.2. First-stage

[Table 2](#) reports the first stage for the term of interest, adding the four excluded instruments one at a time. The final column, which uses the full set, is the specification we estimate.

The first stage is strong. The Sanderson-Windmeijer conditional F is 27.71 in column (1) and 456.42 in column (4), where the full instrument set is included (Sanderson and Windmeijer 2016). Each endogenous term is separately well predicted, not merely well predicted on average, so the high-intensity and post-earthquake interactions are jointly important rather than the direct exchange-rate term carrying the instrument set on its own. [Appendix Table A7](#) confirms the same strength under a wild cluster bootstrap.

Table 2 — First-stage regressions for the remittance TD interaction

	(1)	(2)	(3)	(4)
ER _{hwt}	0.05*** (0.01)	-0.00 (0.00)	-0.08*** (0.01)	-0.00 (0.00)
HI _w x ER _{hwt}		0.08*** (0.02)	0.11*** (0.01)	-0.00 (0.00)
2016-18 x ER _{hwt}			0.10*** (0.01)	-0.00 (0.00)
2016-18 x HI _w x ER _{hwt}				0.14*** (0.01)
SW conditional F	27.71	54.15	219.56	456.42
N	22354	22354	22354	22354
District	71	71	71	71
VDC and year FEs	✓	✓	✓	✓
Ψ _{hwt}	✓	✓	✓	✓

Notes: The dependent variable is the endogenous triple-interaction remittance term. Columns add the excluded exchange-rate instruments one at a time. SW conditional F is the Sanderson-Windmeijer conditional F for that term, computed from the full model in which all four remittance terms are instrumented jointly. Two-stage least squares with VDC and year fixed effects. Earthquake intensity HI_w is measured at the ward, one level below the absorbed unit. X_{hwt} are household controls and Z_w ward characteristics measured once. The exchange-rate instruments enter in levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors, reported in parentheses, are clustered at the district level across 71 districts.

4.3.3. Identifying assumptions

The design layers an instrument on a triple difference: the triple difference compares high- and low-intensity wards before and after the earthquake, while the instrument supplies movement in remittances that households do not themselves choose. Each component carries its own identifying requirement.

The first is the exclusion restriction: conditional on fixed effects and controls, exchange rates affect post-earthquake outcomes only through remittances. Destination-currency movements change the rupee value of a transfer without being chosen by the recipient household (Yang 2008b), and Khanna et al. (2026) use the same logic through pre-existing destination exposure; Nepal’s peg to the Indian rupee leaves households little scope to influence the bilateral variation the instrument uses, as a small open economy with a fixed rate has no monetary autonomy to transmit (Shambaugh 2004; Hull and Imai 2013). The restriction fails if exchange rates instead reach production through local markets. Three features of the setting make that unlikely: our households are smallholders (Table A3) and so are not exposed to output-price movements as sellers on any scale, the identifying variation is at the household-destination-year level rather

than a market-wide price shock, and common macroeconomic effects are absorbed by the time effects and lower-order interactions. We also test it directly. Production valued at fixed national prices, and production measured in physical kilograms, both give the same estimate (Appendix Table A12), and the instrument moves no VDC-year, district-year, or household price index (Appendix Table A13).

The second is a triple-difference parallel-trends condition: absent the earthquake, the remittance–production relationship would have evolved similarly in high- and low-intensity wards (Pathak and Schündeln 2026). The household rounds cannot test it, being repeated cross-sections with only 2011 falling before the earthquake; the year-specific estimates in Panel B of Table 4 run forward from 2011 and therefore speak to persistence rather than to pre-trends. Three other pieces of evidence bear on the condition.

The land-cover design of Section 4.2 tests it directly over the fifteen years before the shock. In Figure 1 the coefficients sit in a flat band slightly above zero from 2003 to 2011, none individually distinguishable from it, and turn negative only after 2015; a linear extrapolation of the pre-earthquake path predicts no decline, so the post-2015 movement is a break rather than a continuation. The same holds with continuous exposure and at the VDC (Appendix C). The test runs on a proxy: cropland is land in use rather than output, and shift-share exposure stands in for remittance potential rather than receipts.

The estimated equation then carries its own counterfactual in γ , the difference-in-differences effect of high intensity evaluated at zero remittances, which describes households that received nothing from abroad; Tables 4 to 8 report it alongside the effect of interest. For these households no production-side margin moves: production, consumption, capital and labor are all indistinguishable from zero, while earthquake aid rises sharply. They were hit, and were reached by the public response, and still show no differential production adjustment; that appears only as remittance inflows rise. Because the γ estimates are imprecise, they are better read as weighing against a large pre-existing divergence than as establishing exact parallelism.

Finally, the selection that motivates instrumenting runs against the result rather than towards it. Its cross-sectional direction is visible in the estimated equation: in the wards later shaken hardest, remittance receipt was associated with *higher* production before the earthquake, significantly so. The triple difference therefore reverses a positive association rather than extending a negative

one.

Two further threats are handled in the appendix. Migrant destination is a household choice measured in the same round as the outcome, so we re-estimate on restricted samples ([Appendix Table A4](#)) and drop one currency bloc or one destination group at a time ([Appendix Figure A2](#)); these condition on a post-treatment variable and are diagnostics rather than alternative estimates. On inference, [Appendix Table A6](#) reports Anderson–Rubin tests valid under weak instruments, its Panel B a wild cluster bootstrap addressing the number and distribution of treated districts, and [Appendix Table A8](#) alternative clustering levels. Robustness to how the outcome is transformed, how intensity is defined, and the geography absorbed is reported with the production estimate in [Section 5](#).

5 — Results

5.1. Farm Production

We begin with output. Every margin the paper goes on to examine — purchased inputs, the allocation of household labor, and the mix of crops and livestock a household continues to produce — operates on how much it produces, so the production response summarizes them and the mechanism sections then ask which of them accounts for it. Burchardi et al. (2019) proceed the same way in estimating the effect of tenancy incentives on inputs and on output.

Production and remittances both carry a large mass at zero: 19.5 percent of household-years report no farm production, and 60.4 percent no remittances. As Chen and Roth (2024) show, the average treatment effect for a log-like transformation defined at zero is then sensitive to the units in which the outcome is measured, because the scaling of the outcome implicitly determines the weight placed on the extensive relative to the intensive margin. That sensitivity is material here, since the unit shift places zeros one rupee below a positive median in the tens of thousands. [Table 3](#) therefore reports the effect under each of the approaches they recommend: columns (1) and (2) retain the conventional log-like transformations for reference, columns (3) and (4) estimate the two margins separately, and columns (5) to (7) calibrate their relative value explicitly. The benchmark applies $\log(y + 1)$ to production, remittances and the instrument alike.

The benchmark coefficient is -0.42 . Away from zero it admits a proportional interpretation:

production falls by roughly four percent for every ten percent rise in remittances. Because zero-producing households remain in the sample, however, this estimate aggregates two distinct margins, a change in the probability of producing at all and a change in output among those that continue, and the weight it places on each is determined implicitly by the units in which production is measured (Chen and Roth 2024).

Columns (5) to (7) therefore calibrate the extensive margin explicitly rather than leaving it to the scaling of the outcome. Production is divided by its smallest positive value before taking logs, and zero-valued observations are assigned $-x$, so that moving from zero to that smallest positive value is valued as the equivalent of a $100x$ log point change along the intensive margin. Across $x = 0.1, 0.5$ and 1 the estimate ranges from -0.42 to -0.46 , so the conclusion is not an artifact of the weight placed on the extensive margin.

Columns (3) and (4) instead estimate the two margins separately. The extensive margin, an indicator for positive production, is -0.04 and precisely estimated; the intensive margin, log output conditional on producing, is -0.03 and cannot be distinguished from zero. The response is thus concentrated at the extensive margin: higher inflows reduce the probability of farming rather than the scale of output among those who continue. Because the two columns are expressed in different units, this ranks the margins by the precision with which each is identified rather than by magnitude, and the confidence interval on column (4) remains consistent with an economically meaningful intensive-margin decline.

Table 4 then decomposes the benchmark production outcome into its constituent categories using the same $\log(y + 1)$ specification. This component decomposition shows whether the total production response reflects a broad-based decline across farm production activities or changes concentrated in a narrower set of crop and livestock categories.

The decline is spread across categories rather than confined to one. Every component is negative, and two can be distinguished from zero: grains at -0.33 and meat at -0.14 . Grains are the largest component of farm production in this setting and depend on purchased seed and fertilizer at planting, while meat requires sustained outlays on animals and feed. Both are therefore categories for which continued production involves ongoing commitments of cash, labor, or working capital. The component pattern is consistent with a post-earthquake reduction in input-intensive production activities. The remaining components are negative but imprecise, and

Table 3 — Total production under alternative outcome transformations

	Dep var: alternative transformations of total production (P)						
	log(P + 1)	lhs(P)	1(P > 0)	log(P) for P > 0	calibrated log(P)		
					x=0.1	x=0.5	x=1
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Panel A. Pooled post-earthquake years 2016–18</i>							
2016-18 x HI_w	0.30 (0.92)	0.34 (0.98)	0.05 (0.08)	-0.18 (0.24)	0.31 (0.93)	0.33 (0.96)	0.35 (1.00)
2016-18 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.42*** (0.13)	-0.44*** (0.14)	-0.04*** (0.01)	-0.03 (0.05)	-0.42*** (0.14)	-0.44*** (0.14)	-0.46*** (0.15)
<i>Panel B. Separate post-earthquake years 2016–18</i>							
2016 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.41*** (0.14)	-0.44*** (0.15)	-0.04*** (0.01)	-0.03 (0.05)	-0.41*** (0.14)	-0.43*** (0.15)	-0.45*** (0.16)
2017 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.46*** (0.14)	-0.49*** (0.15)	-0.05*** (0.01)	-0.00 (0.06)	-0.46*** (0.14)	-0.48*** (0.15)	-0.50*** (0.16)
2018 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.36** (0.14)	-0.38** (0.15)	-0.04** (0.01)	-0.05 (0.06)	-0.36** (0.14)	-0.38** (0.15)	-0.39** (0.16)
N	22354	22354	22354	17999	22354	22354	22354
District	71	71	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓	✓	✓

Notes: Production is the value of output in the seven Food Consumption Score (FCS) food groups the survey records on the production side that predominantly contribute to nutritional intake (WFP 2024). Columns (5)–(7) use calibrated-log production with alternative extensive-margin values x . HI_w indicates above-median ward earthquake intensity, measured by Modified Mercalli Intensity, and $\log(\widehat{R}_{hwt} + 1)$ is the instrumented remittance term, fitted from the matching exchange-rate interactions, which enter in levels. Panel A pools the post-earthquake years 2016–18; Panel B reports the separate post-earthquake year interactions, with 2011 the omitted base year. All specifications are two-stage least squares and include VDC and year fixed effects, Ψ_{hwt} , which collects the three lower-order remittance terms together with the household controls X_{hwt} and the ward controls Z_w defined in the empirical strategy section. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors, reported in parentheses, are clustered at the district level across 71 districts.

none is positive.

The effect is also robust to how output is valued (Appendix Table A12). Production is valued at prices recorded in each household’s local market, so a shift in those prices would register as a change in output even if nothing were grown differently, a concrete possibility given Nepal’s peg to the Indian rupee and the 2015–16 border blockade. Valuing every crop at fixed national prices leaves the estimate close to the benchmark, at -0.37 and -0.42 , and neither a VDC-year nor a district-year price index responds to the instrument (Appendix Table A13). Nor does it depend on

Table 4 – Production components and total

	Dependent variable: log(production variables + 1)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Grains	Pulses	Vegetables	Fruit	Meat	Milk and eggs	Oil	Total
<i>Panel A. Pooled post-earthquake years 2016–18</i>								
2016-18 x HI_w	0.46 (0.84)	-0.40 (0.71)	0.73 (0.91)	0.64 (0.74)	0.81*** (0.26)	-0.62 (0.58)	-0.05 (0.45)	0.30 (0.92)
2016-18 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.33** (0.13)	-0.01 (0.09)	-0.17 (0.11)	-0.14 (0.12)	-0.14*** (0.04)	-0.07 (0.10)	-0.03 (0.07)	-0.42*** (0.13)
<i>Panel B. Separate post-earthquake years 2016–18</i>								
2016 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.31** (0.14)	0.06 (0.11)	-0.12 (0.12)	-0.16 (0.12)	-0.12** (0.05)	-0.13 (0.12)	-0.05 (0.09)	-0.41*** (0.14)
2017 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.36** (0.14)	-0.09 (0.12)	-0.25** (0.13)	-0.14 (0.12)	-0.14*** (0.05)	-0.07 (0.11)	-0.01 (0.08)	-0.46*** (0.14)
2018 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.29** (0.14)	-0.00 (0.10)	-0.12 (0.12)	-0.10 (0.12)	-0.14*** (0.05)	-0.00 (0.13)	-0.04 (0.08)	-0.36** (0.14)
N	22354	22354	22354	22354	22354	22354	22354	22354
District	71	71	71	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Production is the value of output in the seven Food Consumption Score (FCS) food groups the survey records on the production side that predominantly contribute to nutritional intake (WFP 2024). HI_w indicates above-median ward earthquake intensity, measured by Modified Mercalli Intensity, and $\log(\widehat{R}_{hwt} + 1)$ is the instrumented remittance term, fitted from the matching exchange-rate interactions, which enter in levels. Panel A pools the post-earthquake years 2016–18; Panel B reports the separate post-earthquake year interactions, with 2011 the omitted base year. All specifications are two-stage least squares and include VDC and year fixed effects, Ψ_{hwt} , which collects the three lower-order remittance terms together with the household controls X_{hwt} and the ward controls Z_w defined in the empirical strategy section. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors, reported in parentheses, are clustered at the district level across 71 districts.

how the design is specified. The estimate survives the inverse hyperbolic sine and calibrated-log transformations applied to every outcome block (Appendix Table A10), so it is not an artifact of the $\log(y + 1)$ treatment of zeros; replacing the binary intensity indicator with standardized continuous shaking, raising the cutoff above the median, and excluding the wards immediately adjacent to it (Appendix Table A11); absorbing districts rather than VDCs (Appendix Table A9); and dropping the Kathmandu Valley (Appendix Table A14). It stays negative and significant in every case.

5.2. Mechanisms

5.2.1. Consumption Smoothing

Whether the production response documented above is accompanied by a corresponding adjustment in consumption is the first question to put to the mechanism. It is what distinguishes a household that has been made worse off from one that is reallocating, and consumption is the margin on which the insurance role of remittances is conventionally detected.

We measure consumption over the same seven food groups as production, but restricted to what the household purchases. This restriction is essential to the interpretation. In an agricultural household facing thin markets, production and consumption decisions are not separable: what is grown and what is eaten are determined jointly, so the home-produced component of the food budget moves mechanically with output (de Janvry, Fafchamps, and Sadoulet 1991), and in our data it is literally a line item inside the production aggregate. Purchased food is the component least entangled with the production decision, and so the one on which a genuine welfare response would show up rather than an accounting identity. [Table A15](#) reports the same specification on total consumption, including the home-produced component.

[Table 5](#) shows no evidence that households facing a larger remittance response reallocated their consumption portfolio alongside production. The aggregate is indistinguishable from zero, and so is each of the seven food groups. Entering the post-earthquake years separately gives the same result, with one of twenty-four coefficients reaching five percent, as [Appendix Table A18](#) reports.

The consumption interval is too narrow to accommodate an adjustment of the size found in production. On the log scale on which production declines by 0.42, the ninety-five percent interval for the consumption response runs from -0.11 to 0.09 , so a decline of that magnitude is excluded several times over: the null reflects the precision of the estimate rather than a lack of it. Consumption is smoothed while production is not, and it is the size of that wedge — a large production response beside a consumption margin indistinguishable from its pre-shock level — that the remaining mechanisms have to account for.

That fits standard insurance logic in the remittances literature, where transfers respond to adverse shocks in ways that stabilize recipient consumption (Yang and Choi 2007; Combes and

Ebeke 2011), and evidence from Nepal linking remittances to higher food expenditure in normal times (Mishra, Kondratjeva, and Shively 2022). Here the same stabilization appears in a post-disaster setting, with the adjustment falling on farm production instead.

Stable consumption is not evidence that recipient households were uniquely protected. Consumption smoothing is a general motive, and households without remittances have other instruments for it: borrowing, drawing down savings and assets, selling livestock, wage work, and public assistance. Our estimates do not measure those margins, and the design cannot rank them. The estimates do identify remittances as one such instrument, and a sufficient one, for the households that receive them. The evidence is not that recipients were protected and others were not, but that for recipients the protection came from abroad and did not have to come from the farm.

Table 5 – Consumption components and total

	Dependent variable: log(consumption variables + 1)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Grains	Pulses	Vegetables	Fruit	Meat	Milk and eggs	Oil	Total
2016-18 x HI_w	0.90 (0.74)	0.35 (0.68)	-0.65 (0.43)	0.06 (0.57)	0.15 (0.56)	0.19 (0.76)	0.27 (0.42)	0.06 (0.22)
2016-18 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	0.03 (0.17)	-0.18 (0.17)	0.10 (0.09)	-0.14 (0.19)	0.17 (0.18)	-0.07 (0.17)	-0.08 (0.10)	-0.01 (0.05)
N	22354	22354	22354	22354	22354	22354	22354	22354
District	71	71	71	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Consumption is farm output the household buys to eat, excluding home-produced goods that feed directly into the construction of farm production, over the same seven FCS groups as the production outcome (WFP 2024). Table A15 reports the same specification over all household consumption items. The specification, controls, fixed effects and inference are as in Table 4. The triple-difference term pools the post-earthquake years 2016–18; the appendix reports the same specification with each post-earthquake year entered separately.

5.2.2. Capital Inputs

If remittances make it easier to cope without maintaining agricultural activity, the effect should appear first in the purchased inputs that support production: seed, fertilizer, irrigation, equipment rental, handling and upkeep, and livestock inputs. These convert intended production into realized output, and so indicate whether the production result operates through a reduction in production

scale rather than through consumption smoothing alone.

Households with higher remittance inflows in high-intensity areas reduced capital spending on the farm by almost exactly as much as production fell. In [Table 6](#) the capital-input total declines by 0.41 against a production decline of 0.42. The decline is clearest for livestock inputs at -0.37 , for handling and upkeep at -0.20 , and for seed at -0.19 . The remaining components, fertilizer and pesticide, irrigation, and equipment rental, are also negative but cannot be distinguished from zero. That pattern is consistent with a broad but uneven decline in production intensity. Seed and livestock inputs are the two components with the clearest interpretation: both are cash commitments made before output is realized, seed at planting and feed and animals on a continuing basis, so lower spending on them is an input decision rather than a consequence of a poor harvest.

Entering the post-earthquake years separately places the response in time. The capital-input total is -0.51 in 2016 and -0.45 in 2017, both significant, and -0.27 and no longer distinguishable from zero by 2018, as [Appendix Table A19](#) reports. The adjustment is therefore concentrated in the two rounds immediately after the shock.

Table 6 – Capital input components and total

	Dependent variable: $\log(\text{capital input variables} + 1)$						
	(1) Seed	(2) Fertilizer and pesticide	(3) Irrigation	(4) Equipment rental	(5) Handling and upkeep	(6) Livestock	(7) Total
2016-18 x HI_w	0.02 (0.61)	0.18 (0.53)	-0.16 (0.58)	-0.40 (0.74)	0.42 (0.52)	-0.12 (1.04)	0.13 (0.81)
2016-18 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.19* (0.10)	-0.12 (0.10)	-0.05 (0.09)	-0.10 (0.13)	-0.20* (0.10)	-0.37** (0.16)	-0.41** (0.17)
N	22354	22354	22354	22354	22354	22354	22354
District	71	71	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓	✓	✓

Notes: Hired labor is reported with labor inputs and is excluded from the capital total. The specification, controls, fixed effects and inference are as in [Table 4](#). The triple-difference term pools the post-earthquake years 2016–18; the appendix reports the same specification with each post-earthquake year entered separately.

5.2.3. Labor Inputs

If the production effect reflects a genuine reallocation away from farming, labor inputs should weaken too, either because households withdraw their own time or because they hire less outside labor. Table 7 separates the two: spending on hired labor, and the imputed value of self-supplied agricultural labor.

The decline is concentrated entirely in the household's own labor. The value of self-supplied agricultural labor falls by 0.25 and total labor input by 0.26, while spending on hired labor does not fall at all: its coefficient is 0.09 and indistinguishable from zero. Households are not economizing by dismissing hired workers, which is what a pure liquidity contraction would imply; they are withdrawing their own time from agriculture while continuing to pay for outside labor at unchanged rates. Consistent with this, the labor-allocation results in Appendix Table A17 show that household members' wage agricultural labor also declines. That pattern strengthens the interpretation that the production effect operates through lower own agricultural effort alongside the lower purchased inputs documented in Table 6. The asymmetry holds year by year: hired labor is positive and insignificant in each post-earthquake round, at 0.08, 0.12 and 0.09, while the total is negative in all three, as Appendix Table A20 reports.

Table 7 – Labor input components and total

	Dependent variable: log(labor input variables + 1)		
	(1) Hired labor	(2) Own agricultural labor	(3) Total
2016-18 x HI_w	-0.74 (0.76)	-0.75 (0.69)	-0.82 (0.74)
2016-18 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	0.09 (0.11)	-0.25* (0.13)	-0.26** (0.13)
N	22354	22354	22354
District	71	71	71
VDC and year FEs	✓	✓	✓
Ψ_{hwt}	✓	✓	✓

Notes: The specification, controls, fixed effects and inference are as in Table 4. The triple-difference term pools the post-earthquake years 2016–18; the appendix reports the same specification with each post-earthquake year entered separately.

5.2.4. Public Assistance

Public assistance is a coping margin institutionally distinct from private transfers, and it offers a direct test of an alternative explanation for the production result. If recipient households reduced production because public support had already relieved them, or if households without remittances kept producing because no such support arrived, then public assistance should move with remittance inflows. The table reports earthquake aid first, followed by old-age and widow, disability and ethnic, maternal and child, conflict and martyr, other aid, and their total.

Public assistance does not move with remittances. In [Table 8](#), earthquake aid rises by 2.11 log points in high-intensity areas after the shock and total public assistance by 1.73, both significant at the five percent level. Because most households received no aid before the earthquake, these are best read as aid appearing where there was none rather than as proportional increases. Because that term is evaluated where $\log(R_{hwt} + 1) = 0$, it describes households with no remittance income. The triple-difference term asks how that response varies with remittance inflows: for earthquake aid it is small and negative, and no component or total can be distinguished from zero. Entering the post-earthquake years separately leaves the same picture, with a small number of coefficients significant at ten percent in single years and none surviving pooling, as [Appendix Table A21](#) reports.

At the margin, an additional log point of remittances moves earthquake aid by at most a fifth of the amount by which it moves production. The ninety-five percent interval runs from -0.09 to 0.03 , so the data admit a small increase as readily as a small decrease and exclude anything close to the production response. Read against the production estimates, this locates the two coping margins. Aid arrives in high-intensity areas whether or not a household receives transfers from abroad, while production falls only as inflows rise. Differential access to public support therefore cannot account for the production decline, and remittances operate alongside public assistance rather than in place of it. We do not describe this as crowding out of public transfers: that would require the triple-difference estimate to be negative and distinguishable from zero, and it is not. What remittances crowd out is farm production, not public support.

Table 8 – Public assistance components and total

	Dependent variable: log(public assistance + 1)						
	(1) Earthquake aid	(2) Old age and widow	(3) Disability and ethnic	(4) Maternal and child	(5) Conflict and martyr	(6) Other aid	(7) Total
2016-18 x HI_w	2.11** (0.87)	0.06 (0.30)	-0.03 (0.09)	-0.18 (0.11)	-0.01 (0.01)	-0.01* (0.01)	1.73** (0.76)
2016-18 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.03 (0.03)	-0.12 (0.11)	-0.00 (0.02)	0.05 (0.03)	0.00 (0.00)	0.00 (0.00)	-0.09 (0.11)
N	22354	22354	22354	22354	22354	22354	22354
District	71	71	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓	✓	✓

Notes: The specification, controls, fixed effects and inference are as in Table 4. The triple-difference term pools the post-earthquake years 2016–18; the appendix reports the same specification with each post-earthquake year entered separately.

5.3. Robustness

5.4. Robustness

Section 4.3.3 sets out the threats to the design and where each is tested. All apply to the mechanism outcomes as well as to production, and the pattern across margins is broadly unchanged, with the labor estimate the most sensitive. The consumption and public-assistance nulls are the most stable feature: Anderson–Rubin fails to reject the joint null for either, and their wild cluster bootstrap p -values stay large. Appendix Table A4 shows capital remaining negative and significant as the sample is narrowed, while labor weakens under the narrowest restrictions. Appendix Table A6 points the same way: Anderson–Rubin tests reject the joint null for production, capital and labor, though the confidence set projected onto the labor coefficient includes zero, and the wild cluster bootstrap leaves production and capital significant while moving labor to the ten percent margin. We therefore read the capital response as firm and the labor response as suggestive, and rest the argument on the pattern across margins rather than on the labor estimate alone.

6 — Discussion

6.1. Interpreting the Production Response

The three questions set out in [Section 1](#) can now be answered. The remittance response to the earthquake did not preserve farm production; it displaced it. The adjustment fell on farm inputs and household labor rather than on food consumption. And it did not operate through differential access to public support: aid rose in affected areas, and higher remittance inflows did not reduce it.

The response is concentrated at the extensive margin: what changes is whether a household farms at all, not how intensively it farms conditional on farming. That distinction matters for persistence: a household that reduces fertilizer can restore it the following season, whereas one that has let land preparation lapse, sold animals and forgone planting faces re-entry costs. The timing of the capital response reported above, concentrated in the first two post-earthquake rounds and absent in the third, is what that account predicts.

The gap between the production and consumption responses is the second feature that shapes the reading. Production falls substantially while the consumption estimate is small and precisely estimated around zero, tight enough to exclude an adjustment of the size observed in production several times over. Households are not absorbing the shock by eating less. This is the pattern the mechanisms have to account for, and it is what distinguishes a household made worse off from one reallocating away from a margin it is no longer compelled to hold.

6.2. Accounting for the Production Decline

The reduced-form estimates identify the response of each margin separately but do not establish how much of the production decline each accounts for. [Table 9](#) addresses that question directly, decomposing the triple-difference IV effect one mediator at a time through consumption, capital inputs, labor inputs, and public assistance, following Imai, Keele, and Tingley (2010).

A causal reading of such a decomposition requires sequential ignorability: each mediator must be as good as random with respect to potential production once treatment, controls, and fixed effects are conditioned on. That assumption is demanding here, because consumption, capital use, labor allocation and public assistance are themselves post-earthquake choices that move with

unobserved recovery needs and local conditions. We take it to be violated rather than merely untestable, and therefore do not read the shares as causal channel weights.

Read as association, the decomposition returns the same ordering as the separate estimates. Consumption accounts for none of the production effect: the share running through it is small, negative and indistinguishable from zero. The indirect effects through capital and labor are large, at roughly 71 and 44 percent of the total. Public assistance accounts for none of it. Because the shares are estimated one mediator at a time and are not constrained to sum to the total, they measure the relative importance of each margin rather than an exhaustive decomposition. [Appendix Table A16](#) repeats the exercise with consumption measured over all household items, and [Appendix B](#) sets out the estimating equations.

Two procedures with different failure modes return the same ranking of margins: separate reduced-form estimates under the triple-difference IV assumptions, and a decomposition under sequential ignorability. Because the two would not fail in the same direction, that agreement is more persuasive than either alone — but the argument rests on the separately estimated margins, not on the shares.

Table 9 — IV mediation of total production through transformed totals

	Dependent variable: log(total production + 1)				
	Farm production	Consumption smoothing	Capital inputs	Labor inputs	Public assistance
Total effect (TD)	-0.42*** (0.14)	-0.42*** (0.14)	-0.42*** (0.14)	-0.42*** (0.14)	-0.42*** (0.14)
Indirect effect		0.00 (0.02)	-0.30*** (0.16)	-0.18*** (0.11)	-0.00 (0.00)
Direct effect		-0.42*** (0.13)	-0.12 (0.08)	-0.23*** (0.08)	-0.42*** (0.14)
Proportion mediated		-0.37 (3.20)	71.05*** (22.49)	44.39*** (17.54)	0.03 (0.22)
N	22354	22354	22354	22354	22354

Notes: The outcome is log(total farm production + 1), summed over the seven FCS food groups the survey records on the production side (WFP 2024). Every column is a household total. Farm production is the total effect with no mediator; the remaining columns decompose it through one mediator at a time. Consumption smoothing is the purchased-food measure of Table 5. Capital input is spending on seed, fertilizer and pesticide, irrigation, equipment rental, handling and upkeep, and livestock inputs. Labor input is spending on hired agricultural labor plus the imputed value of the household’s own agricultural labor. Public assistance is total government and program transfers. Standard errors are bootstrap standard deviations across 50 district-clustered replications, and stars follow two-sided sign-based bootstrap p-values. Proportion mediated is in percent and is unstable when the total effect is near zero in bootstrap draws. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The bootstrap resamples across 71 districts.

6.3. Implications for Post-Disaster Policy

Remittances help households cope with the shock, but not by sustaining pre-earthquake production, capital or input use. They ease short-run budget constraints, making it more feasible to scale back farming when sustaining it is no longer the binding margin of adjustment. Households without remittance income meet the same shock without that slack and lean on public support instead.

Two implications follow for how recovery is supported. First, private transfers are not a substitute for policies aimed at sustaining production after a disaster. By easing the budget constraint they may reduce the pressure to sustain it, so a recovery strategy that treats inbound remittances as a substitute for agricultural support will understate what is needed. Second, the margins on which production falls identify where intervention would operate: purchased inputs, farm labor, and short-run working capital, rather than consumption support. That aid did not fall as remittance inflows rose indicates the gap is not one that existing relief instruments close on their own.

Lower production alongside stable consumption is not evidence that recipient households were made worse off: it is equally consistent with households reallocating away from a low-return activity once an outside income source removes the compulsion to hold it. What the estimates do establish is that the aggregate agricultural recovery of affected areas was slower than it would have been absent the remittance response, which is a different question from household welfare and the one relevant to agricultural policy.

6.4. Limitations

Four limitations bound what these estimates establish. First, the design identifies the effect for households whose transfers moved with the exchange rate, so it speaks to marginal receipts rather than to remittances as such. Second, the surveys contain a single pre-earthquake round, so the parallel-trends condition rests on the land-cover evidence and on the absence of any production response among non-remitting households rather than on a pre-trend in the estimating sample itself. Third, the mediation decomposition requires an assumption we do not think holds here, and so corroborates the separately estimated margins rather than standing on its own. Fourth, the horizon is short: we observe three post-earthquake years, and whether the production gap closes

as recovery proceeds is a question these data cannot answer.

7 — Conclusion

Whether private transfers sustain production after a disaster, or relieve households of the need to sustain it, matters for how recovery is financed and for the role left to public support. The question has been hard to answer because remittances respond to the same shocks that affect production.

Using exchange-rate-driven variation in remittances within a triple-difference IV design, we find that higher remittance inflows after the Nepal earthquake reduced farm production in the short run, and that the response operates at the extensive margin: households leave farming rather than farm less intensively. Consumption does not fall, neither in the aggregate nor in any of its seven food groups, while capital inputs and the household's own agricultural labor decline. The component results point to the categories requiring continued cash, labor or working-capital commitments: grains and meat on the output side, livestock inputs, handling and upkeep, and seed on the input side. Earthquake aid rises sharply in affected areas and does not fall as inflows rise, so the transfers displace farm production while leaving public support intact.

For recovery policy the implication is direct: inbound remittances are not a substitute for support aimed at sustaining production. Because they ease the budget constraint rather than fund the farm, a recovery strategy that counts on them to carry agricultural activity will fall short, and the margins on which production contracts — purchased inputs, farm labor, and short-run working capital — are where such support would have to act.

The broader pattern is one of post-disaster adjustment in which remittances relax the need to maintain farm production at pre-shock levels. A longer panel would show whether that reallocation persists and whether the households that left farming return to it as recovery proceeds.

References

- Adão, Rodrigo, Michal Kolesár, and Eduardo Morales.** 2019. “Shift-Share Designs: Theory and Inference.” *The Quarterly Journal of Economics* 134 (4): 1949–2010. <https://doi.org/10.1093/qje/qjz025>. (Cited on page 72).
- Anderson, T. W., and Herman Rubin.** 1949. “Estimation of the Parameters of a Single Equation in a Complete System of Stochastic Equations.” *The Annals of Mathematical Statistics* 20 (1): 46–63. <https://doi.org/10.1214/aoms/1177730090>. (Cited on pages 53, 54).
- Borusyak, Kirill, Peter Hull, and Xavier Jaravel.** 2022. “Quasi-Experimental Shift-Share Research Designs.” *The Review of Economic Studies* 89 (1): 181–213. <https://doi.org/10.1093/restud/rdab030>. (Cited on pages 10, 13, 72).
- . 2025. “A Practical Guide to Shift-Share Instruments.” *Journal of Economic Perspectives* 39 (1): 181–204. <https://doi.org/10.1257/jep.20231370>. (Cited on pages 13, 72).
- Burchardi, Konrad B, Selim Gulesci, Benedetta Lerva, and Munshi Sulaiman.** 2019. “Moral Hazard: Experimental Evidence from Tenancy Contracts.” *The Quarterly Journal of Economics* 134 (1): 281–347. <https://doi.org/10.1093/qje/qjy023>. (Cited on page 18).
- Central Bureau of Statistics.** 2001. *National Population Census 2001, Tenth Census [Data set]*. Study NPL-CBS-NPC-2001-V2. National Data Archive, National Statistics Office, Nepal. Accessed August 27, 2026. <https://microdata.nsonepal.gov.np/index.php/catalog/42>. (Cited on pages 9, 71).
- . 2011a. *National Population and Housing Census 2011 [Data set]*. Study NPL-CBS-NPHC-2011-V1. National Data Archive, National Statistics Office, Nepal. Accessed August 27, 2026. <https://microdata.nsonepal.gov.np/index.php/catalog/54>. (Cited on pages 2, 9).
- . 2011b. *Nepal Living Standards Survey 2010-2011, NLSS Third [Data set]*. National Data Archive, National Statistics Office, Nepal. Accessed August 27, 2026. <https://microdata.nsonepal.gov.np/index.php/catalog/37>. (Cited on pages 2, 7).

- Chen, Jiafeng, and Jonathan Roth.** 2024. “Logs with Zeros? Some Problems and Solutions.” *The Quarterly Journal of Economics* 139 (2): 891–936. <https://doi.org/10.1093/qje/qjad054>. (Cited on pages 4, 18, 19, 59).
- Combes, Jean-Louis, and Christian Ebeke.** 2011. “Remittances and Household Consumption Instability in Developing Countries.” *World Development* 39, no. 7 (July): 1076–1089. ISSN: 0305-750X. <https://doi.org/10.1016/j.worlddev.2010.10.006>. (Cited on pages 5, 22).
- Davidson, Russell, and James G. MacKinnon.** 2010. “Wild Bootstrap Tests for IV Regression.” *Journal of Business & Economic Statistics* 28 (1): 128–144. <https://doi.org/10.1198/jbes.2009.07221>. (Cited on pages 54, 55).
- de Janvry, Alain, Marcel Fafchamps, and Elisabeth Sadoulet.** 1991. “Peasant Household Behaviour with Missing Markets: Some Paradoxes Explained.” *The Economic Journal* 101 (409): 1400–1417. <https://doi.org/10.2307/2234892>. (Cited on pages 22, 64).
- Duflo, Esther, Michael Kremer, and Jonathan Robinson.** 2008. “How High Are Rates of Return to Fertilizer? Evidence from Field Experiments in Kenya.” *American Economic Review* 98 (2): 482–488. <https://doi.org/10.1257/aer.98.2.482>. (Cited on page 5).
- Ebeke, Christian, and Jean-Louis Combes.** 2013. “Do Remittances Dampen the Effect of Natural Disasters on Output Growth Volatility in Developing Countries?” *Applied Economics* 45 (16): 2241–2254. <https://doi.org/10.1080/00036846.2012.659347>. (Cited on page 5).
- Eldemerdash, Nadia, and Steven T. Landis.** 2023. “The Divergent Effects of Remittance Transfers for Post-Disaster States.” *Economics of Disasters and Climate Change* 7, no. 3 (November): 483–501. ISSN: 2511-1299. <https://doi.org/10.1007/s41885-023-00136-y>. (Cited on page 5).
- Fafchamps, Marcel, Christopher Udry, and Katherine Czukas.** 1998. “Drought and Saving in West Africa: Are Livestock a Buffer Stock?” *Journal of Development Economics* 55 (2): 273–305. ISSN: 0304-3878. [https://doi.org/10.1016/S0304-3878\(98\)00037-6](https://doi.org/10.1016/S0304-3878(98)00037-6). (Cited on page 5).

- Gallagher, Justin, and Daniel Hartley.** 2017. “Household Finance after a Natural Disaster: The Case of Hurricane Katrina.” *American Economic Journal: Economic Policy* 9, no. 3 (August): 199–228. <https://doi.org/10.1257/pol.20140273>. (Cited on page 5).
- Goda, Katsuichiro, Takashi Kiyota, Rama Mohan Pokhrel, Gabriele Chiaro, Toshihiko Katagiri, Keshab Sharma, and Sean Wilkinson.** 2015. “The 2015 Gorkha Nepal Earthquake: Insights from Earthquake Damage Survey.” *Frontiers in Built Environment* 1:8. <https://doi.org/10.3389/fbuil.2015.00008>. (Cited on page 6).
- Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift.** 2020. “Bartik Instruments: What, When, Why, and How.” *American Economic Review* 110 (8): 2586–2624. <https://doi.org/10.1257/aer.20181047>. (Cited on pages 13, 71).
- Government of Nepal, National Planning Commission.** 2015. *Nepal Earthquake 2015: Post Disaster Needs Assessment*. Vols. A and B. Volume B available at https://www.preventionweb.net/files/44973_pdnvolumeb.pdf. Kathmandu: Government of Nepal, National Planning Commission. Accessed August 27, 2026. https://www.preventionweb.net/files/44973_lv9argo0el2dmpnt3gv20170223.pdf. (Cited on page 3).
- Gubert, Flore.** 2002. “Do Migrants Insure Those Who Stay Behind? Evidence from the Kayes Area (Western Mali).” *Oxford Development Studies* 30 (3): 267–287. <https://doi.org/10.1080/1360081022000012699>. (Cited on page 5).
- Guha-Sapir, Debarati, Philippe Hoyois, and Regina Below.** 2017. *Annual Disaster Statistical Review 2015: The Numbers and Trends*. Technical report. Brussels, Belgium: Centre for Research on the Epidemiology of Disasters (CRED), Institute of Health and Society (IRSS), Université catholique de Louvain. Accessed August 28, 2026. <https://www.preventionweb.net/publication/annual-disaster-statistical-review-2015-numbers-and-trends>. (Cited on page 2).
- Hull, Peter, and Masami Imai.** 2013. “Economic Shocks and Civil Conflict: Evidence from Foreign Interest Rate Movements.” *Journal of Development Economics* 103:77–89. <https://doi.org/10.1016/j.jdeveco.2013.02.001>. (Cited on page 16).

- Imai, Kosuke, Luke Keele, and Dustin Tingley.** 2010. “A General Approach to Causal Mediation Analysis.” *Psychological Methods* 15 (4): 309. <https://doi.org/10.1037/a0020761>. (Cited on pages 6, 28, 70).
- IPCC.** 2014. *Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. [Core Writing Team, R.K. Pachauri / L.A. Meyer (eds.)]. IPCC, Geneva, Switzerland, pp., 151. Accessed August 28, 2026. <https://www.ipcc.ch/report/ar5/syr/>. (Cited on page 2).
- Karlan, Dean, Robert Osei, Isaac Osei-Akoto, and Christopher Udry.** 2014. “Agricultural Decisions after Relaxing Credit and Risk Constraints.” *The Quarterly Journal of Economics* 129 (2): 597–652. <https://doi.org/10.1093/qje/qju002>. (Cited on page 5).
- Khanna, Gaurav, Emir Murathanoglu, Caroline B. Theoharides, and Dean Yang.** 2026. “Abundance from Abroad: Migrant Income and Long-Run Economic Development.” *American Economic Review* 116 (4): 1540–1577. <https://doi.org/10.1257/aer.20241465>. (Cited on pages 2, 9, 16, 71, 72).
- Klomp, Jeroen, and Barry Hoogezand.** 2018. “Natural Disasters and Agricultural Protection: A Panel Data Analysis.” *World Development* 104:404–417. ISSN: 0305-750X. <https://doi.org/10.1016/j.worlddev.2017.11.013>. (Cited on page 5).
- Lokshin, Michael, and Elena Glinskaya.** 2009. “The Effect of Male Migration on Employment Patterns of Women in Nepal.” *The World Bank Economic Review* 23 (3): 481–507. ISSN: 02586770, 1564698X. <https://doi.org/10.1093/wber/lhp011>. (Cited on page 4).
- MacKinnon, James G., and Matthew D. Webb.** 2018. “The Wild Bootstrap for Few (Treated) Clusters.” *The Econometrics Journal* 21 (2): 114–135. <https://doi.org/10.1111/ectj.12107>. (Cited on pages 53, 55).
- McAuliffe, Marie, and Anna Triandafyllidou.** 2021. *World Migration Report 2022*. Technical report. Geneva: International Organization for Migration (IOM). Accessed August 28, 2026. <https://publications.iom.int/books/world-migration-report-2022>. (Cited on page 2).

- Mishra, Khushbu, Olga Kondratjeva, and Gerald E. Shively.** 2022. “Do Remittances Reshape Household Expenditures? Evidence from Nepal.” *World Development* 157:105926. ISSN: 0305-750X. <https://doi.org/10.1016/j.worlddev.2022.105926>. (Cited on pages 2, 23).
- Mobarak, Ahmed Mushfiq, Iffath Sharif, and Maheshwor Shrestha.** 2023. “Returns to International Migration: Evidence from a Bangladesh-Malaysia Visa Lottery.” *American Economic Journal: Applied Economics* 15, no. 4 (October): 353–388. ISSN: 1945-7782. <https://doi.org/10.1257/app.20220258>. (Cited on pages 2, 3).
- National Statistics Office.** 2023. *National Population and Housing Census 2021, 12th Population Census [Data set]*. Study NPL-NSO-NPHC-2021-V01. National Data Archive, National Statistics Office, Nepal. Accessed August 27, 2026. <https://microdata.nsonepal.gov.np/index.php/catalog/124>. (Cited on page 2).
- NLCMS.** 2022. *National Land Cover Monitoring System of Nepal (NLCMS)*. Technical report. Government of Nepal, Ministry of Forests, Environment, Forest Research, and Training Center. Accessed August 27, 2026. <https://servir.icimod.org/science-applications/national-land-cover-monitoring-system-nepal/>. (Cited on pages 9, 11, 75).
- Pathak, Prakash, and Matthias Schündeln.** 2022. “Social Hierarchies and the Allocation of Development Aid: Evidence from the 2015 Earthquake in Nepal.” *Journal of Public Economics* 209:104607. <https://doi.org/10.1016/j.jpubeco.2022.104607>. (Cited on page 6).
- . 2026. “Institutions under Pressure: The Effect of Community Groups on Forest Preservation after a Natural Disaster.” *The Economic Journal*, ueag003. <https://doi.org/10.1093/ej/ueag003>. (Cited on pages 6, 17, 60).
- Phadera, Lokendra.** 2016. *International Migration and its Effect on Labor Supply of the Left-Behind Household Members: Evidence from Nepal*. 2016 Annual Meeting, July 31-August 2, Boston, Massachusetts 235968. Agricultural and Applied Economics Association, July. <https://doi.org/10.22004/ag.econ.235968>. (Cited on page 4).

- Ratha, Dilip, Vandana Chandra, Eung Ju Kim, Sonia Plaza, and Akhtar Mahmood.** 2024. *Remittances Slowed in 2023, Expected to Grow Faster in 2024*. Migration and Development Brief 40. Washington, DC: World Bank. Accessed August 27, 2026. <https://www.worldbank.org/en/brief/2024/09/18/remittances-knomad>. (Cited on page 2).
- Sanderson, Eleanor, and Frank Windmeijer.** 2016. “A Weak Instrument F-Test in Linear IV Models with Multiple Endogenous Variables.” *Journal of Econometrics* 190 (2): 212–221. <https://doi.org/10.1016/j.jeconom.2015.06.004>. (Cited on page 15).
- Sapkota, Chandan.** 2013. “Remittances in Nepal: Boon or Bane?” *The Journal of Development Studies* 49 (10): 1316–1331. <https://doi.org/10.1080/00220388.2013.812196>. (Cited on page 6).
- Seddon, David, Jagannath Adhikari, and Ganesh Gurung.** 2002. “Foreign Labor Migration and the Remittance Economy of Nepal.” *Critical Asian Studies* 34 (1): 19–40. <https://doi.org/10.1080/146727102760166581>. (Cited on page 6).
- Shambaugh, Jay C.** 2004. “The Effect of Fixed Exchange Rates on Monetary Policy.” *The Quarterly Journal of Economics* 119 (1): 301–352. <https://doi.org/10.1162/003355304772839605>. (Cited on page 16).
- Shrestha, Maheshwor.** 2019. “Death Scares: How Potential Work-Migrants Infer Mortality Rates from Migrant Deaths.” *Journal of Development Economics* 141:102368. ISSN: 0304-3878. <https://doi.org/10.1016/j.jdeveco.2019.07.001>. (Cited on pages 3, 6).
- Theoharides, Caroline.** 2018. “Manila to Malaysia, Quezon to Qatar: International Migration and Its Effects on Origin-Country Human Capital.” *Journal of Human Resources* 53 (4): 1022–1049. ISSN: 0022-166X, 1548-8004. <https://doi.org/10.3368/jhr.53.4.0216.7714R1>. (Cited on page 3).
- USGS.** 2017. *Advanced National Seismic System (ANSS) Comprehensive Catalog of Earthquake Events and Products*. U.S. Geological Survey, Earthquake Hazards Program. <https://doi.org/10.5066/F7MS3QZH>. (Cited on page 6).

- World Bank.** 2011. *Large-Scale Migration and Remittance in Nepal: Issues, Challenges and Opportunities*. Report 55390-NP. World Bank, South Asia Region, Poverty Reduction and Economic Management. Accessed August 28, 2026. http://www.esocialsciences.org/Articles/show_Article.aspx?acat=InstitutionalPapers&aid=11251. (Cited on page 6).
- . 2020. *Household Risk and Vulnerability Survey, Full Panel 2016-2018 [Data set]*. World Bank, Development Data Group. <https://doi.org/10.48529/mqys-cp36>. (Cited on page 7).
- World Food Programme.** 2024. *Food Consumption Score & Food Consumption Score Nutritional Analysis: Guidance Note*. Technical report. Replaces the 2008 Food Consumption Score and 2015 FCS-N guidance. Rome: World Food Programme, Vulnerability Analysis and Mapping, April. Accessed August 28, 2026. <https://vamresources.manuals.wfp.org/docs/food-consumption-score>. (Cited on pages 8, 20, 21, 23, 29, 41, 66).
- Yang, Dean.** 2008a. “Coping with Disaster: The Impact of Hurricanes on International Financial Flows, 1970-2002.” Cited by: 183; All Open Access, Green Open Access, *B.E. Journal of Economic Analysis and Policy* 8 (1). <https://doi.org/10.2202/1935-1682.1903>. (Cited on pages 3, 5).
- . 2008b. “International Migration, Remittances and Household Investment: Evidence from Philippine Migrants’ Exchange Rate Shocks*.” *The Economic Journal* 118 (528): 591–630. ISSN: 1468-0297. <https://doi.org/10.1111/j.1468-0297.2008.02134.x>. (Cited on pages 2, 3, 7, 9, 15, 16, 42, 44, 45, 51, 52, 72).
- Yang, Dean, and HwaJung Choi.** 2007. “Are Remittances Insurance? Evidence from Rainfall Shocks in the Philippines.” *The World Bank Economic Review* 21, no. 2 (May): 219–248. ISSN: 0258-6770. <https://doi.org/10.1093/wber/lhm003>. (Cited on pages 3, 5, 22).

Appendix

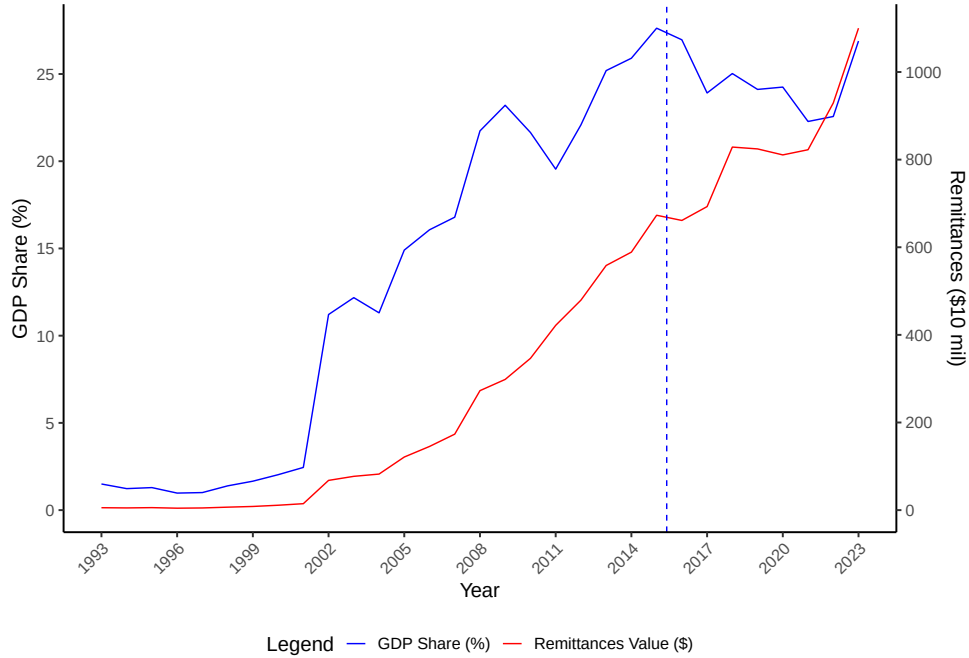
A — Data, Identification, Inference, and Robustness

This appendix has five parts. [Appendix A.1](#) documents the data, [Appendix A.2](#) the identifying variation, [Appendix A.3](#) inference and the levels at which the design operates, [Appendix A.4](#) robustness, and [Appendix A.5](#) results that extend the main tables rather than testing them.

Three conventions hold throughout. Every table that reports outcomes reports the same five — farm production, consumption smoothing, capital inputs, labor inputs and public assistance — in that order, so that a reader compares down a column. Every table reports the pooled 2016–18 interaction, because the year-by-year estimates have a subsection of their own. And every table that varies something places each variation in its own column with the benchmark first. On notation, [Table A4](#) states the estimating specification in full; later tables carry what is specific to them and refer back to it rather than repeating it.

[Figure A1](#) gives the aggregate backdrop. Remittances rose from a negligible share of Nepali GDP in the early 1990s to more than a fifth by the end of the period, which is what makes the household dependence the paper estimates a national phenomenon rather than a local one, and what makes the exchange rate a variable of first-order importance for rural incomes.

Figure A1 – Remittances Relative to GDP Over Time



Notes: Remittance receipts as a share of GDP in Nepal. Source: Authors' calculations.

A.1. Data, sample, and measurement

Each variable is defined below by the survey module it is built from, so that every aggregate the paper uses can be traced to its source. The descriptive comparison between receiving and non-receiving households, the construction of the household exchange rate, and the distribution of production across the sample follow.

A.1.1. Variable construction

Production. Our main outcome is farm production, the value of a household's crop and livestock output over the survey year. The two come from different modules. Crop production is built from the farming module, which records for each crop the quantity the household harvested over the past twelve months and the unit price at which it sold, so output is valued at the prices prevailing in the household's own local market. Livestock production comes from the livestock module, which records output by product: milk, ghee, eggs, curd, meat and other produce. It is the value of output rather than of sales, so produce consumed inside the household still counts. Total production is the value of output in the seven Food Consumption Score (FCS) food groups

the survey records on the production side — grains, pulses, vegetables, fruit, meat, milk and eggs, and oil — which are the groups that predominantly contribute to nutritional intake (WFP 2024). We convert all monetary variables in the paper to 2018 Nepalese rupees (NPR) using the national consumer price index, so the baseline round and the three post-earthquake rounds are comparable.⁵

Consumption, farm inputs, and public assistance. We build three further outcome blocks the same way, as sums of their survey components. Consumption comes from the food module, which is what makes the separation from production possible: for each item the survey records the quantity the household consumed and splits it by source, into food it produced at home, food it purchased in the market, and food received in kind, each valued at market rates. We take the purchased component, over the same seven FCS groups as farm production, so the two aggregates are defined over identical foods and the comparison between them is meaningful, for the reason given in [Section 5.2.1](#).⁶ We restrict consumption to food for two reasons: it is the only consumption component the 2011 and the 2016–18 instruments measure on a comparable basis, and it is the margin on which households are most likely to absorb a transitory shock. Capital inputs cover the cash-intensive agricultural inputs: seeds, fertilizer, irrigation, rented equipment, handling and upkeep, and livestock inputs. Labor inputs cover spending on hired agricultural labor and the imputed value of the household’s own agricultural labor. We also report total labor hours, which sum wage and self-employed hours inside and outside agriculture, in [Table A1](#) and [Table A17](#). Public assistance covers government and program transfers, including earthquake relief, the old-age and widow allowance, disability and ethnic-minority allowances, maternal and child grants and conflict-related payments.

Remittances. Remittances are the amount a migrant sent to the household over the survey year, which the household reports separately for each migrant. We sum over the household’s migrants, so the measure is the household’s total receipts and is zero for households with no migrant. A

5. The rupee traded at roughly NPR 110 to the US dollar in 2018, so the figures below convert at close to \$1 per NPR 100. The rupee has since depreciated, and a current conversion would understate the real values reported here.

6. Items that carry no nutritional weight in the Food Consumption Score are excluded, as is food eaten outside the home, which the two surveys record on different bases. [Table A15](#) reports the same specification over all household consumption items.

substantial share of households receive nothing, so the benchmark specification enters remittances as $\log(1 + R)$, which keeps those households in the estimating sample. [Appendix A.4.1](#) reports the inverse hyperbolic sine and calibrated-log alternatives.

Earthquake intensity. Earthquake intensity is the Modified Mercalli Intensity (MMI) at the household’s ward, taken from the USGS ShakeMap for the 25 April 2015 event. MMI measures the ground shaking actually experienced at a location rather than the magnitude of the earthquake at its source, so it captures exposure rather than distance to the epicenter. We classify a ward as high intensity if its MMI lies above the sample median, which is 5.2. One detail of the ShakeMap data matters for reading the cutoff. MMI is not continuous in the released grid: it is recorded in increments of 0.2, so no ward sits between 5.2 and 5.4. A rule of “above 5.2” therefore admits its first ward at 5.4, and that is the value at which the treated group actually begins. We state the cutoff at 5.4 wherever the exact threshold matters, as in [Table A11](#), and at the median elsewhere; the two describe the same partition of wards. [Figure A3](#) in [Appendix A](#) maps ward-level intensity alongside district-average remittances and production, and shows that the surveys cover the shaken districts and the rest of the country alike rather than concentrating in either.

Exchange rates. The exchange-rate measure is the rate between the Nepalese rupee and the currency of the migrant’s destination country, averaged over the year before the household’s interview date. It varies across households through the destinations they send migrants to, and over time through currency movements and the interview date. For a household with more than one migrant we average across its migrants, the migrant-count weighting of Yang (2008b); that rule binds for only 3.3 percent of the estimation sample. [Appendix A](#) gives the sources, the two coding conventions for internal and non-migrant households, and the aggregation statistics, and [Table A5](#) reports the results under two alternative rules.

Controls. Household controls X_{hwt} are measured in each round and cover the ethnicity, age group and marital status of the head. Household size is excluded for the reason given in [Section 4.3](#). Ward controls Z_w are fixed at their 2011 census values and cover ward area, the poverty index, population, and the shares of agricultural months and agricultural occupations; they carry no time subscript because they are measured once. Rainfall is the exception among the ward variables: it

is averaged over the twelve months before each household’s interview date, so it varies within a ward over time, and we group it with the household controls. The main effects of HI_w and 2016–18 are absorbed by the fixed effects.

A.1.2. Descriptive statistics

Table A1 reports means for the main household characteristics and for the outcome blocks we estimate, splitting households by whether they received any remittances during the survey year. About forty percent of household-years receive remittances. Receiving households are smaller, 4.8 against 5.0 members, and far more likely to be headed by a woman, 39 against 13 percent, which is the expected composition effect of adult male out-migration. They are also more likely to be high-caste, 16 against 10 percent, and married, 69 against 64 percent, and similar in the age of the head. The two groups face the same earthquake intensity on average, an MMI of 5.14 against 5.19, so the remittance split does not stand in for exposure to the earthquake.

Receiving and non-receiving households are statistically indistinguishable in total production, total food consumption and total capital inputs. Two margins do differ. Receiving households supply substantially fewer household labor hours, 2,595 against 3,262 a year, and they receive less public assistance, NPR 8,400 against NPR 10,900. Neither gap is causal, because receiving and non-receiving households differ on many margins at once, which is what the triple-difference design in Section 4.1 addresses. They do locate where the two groups differ most before any of that structure is imposed: labor and public transfers, not production or consumption. The sample is small-scale throughout, with median annual production of NPR 31,000 and three quarters of household-years producing less than NPR 70,000 (Table A3).

Table A1 – Descriptive statistics: remittance-receiving and non-receiving households

	Full sample	Non- receiving	Receiving	Difference	P-value
<i>Panel A. Household characteristics</i>					
Head is high caste	0.12	0.10	0.16	0.06	0.00
Head is female	0.23	0.13	0.39	0.26	0.00
Age of head (years)	48.74	48.54	49.06	0.53	0.10
Head is married	0.66	0.64	0.69	0.06	0.00
Household size (members)	4.93	5.01	4.80	-0.21	0.00
<i>Panel B. Ward characteristics</i>					
Ward area (sq. km)	4.28	4.23	4.35	0.12	0.60
Ward population (2011 census)	378	366	396	30	0.10
Ward poverty index (0–1)	0.28	0.28	0.27	-0.01	0.04
Agricultural months (0–12)	5.59	5.60	5.57	-0.02	0.81
Agricultural occupation share (0–1)	0.73	0.72	0.73	0.01	0.60
Rainfall, 12-month mean (mm)	136	135	137	2	0.35
<i>Panel C. Treatment and instrument</i>					
Earthquake intensity (MMI)	5.17	5.19	5.14	-0.05	0.44
High-intensity ward (share)	0.49	0.49	0.48	-0.01	0.55
Remittances received (in '000)	72.56	0.00	185.61	185.61	0.00
Exchange rate (NPR per unit)	5.27	0.99	11.95	10.96	0.00
<i>Panel D. Outcomes</i>					
Farm production (in '000)	79.79	68.48	97.42	28.95	0.47
Consumption (in '000)	63.02	63.81	61.79	-2.03	0.15
Capital inputs (in '000)	11.09	10.76	11.61	0.85	0.13
Labor inputs, value (in '000)	99.21	99.90	98.15	-1.75	0.53
Labor inputs (hours per year)	3001	3262	2595	-667	0.00
Public assistance (in '000)	9.91	10.90	8.37	-2.53	0.00
Household-years	22354	13615	8739		

Notes: One observation per household-year, on the estimation sample. Monetary amounts are in thousands of 2018 Nepalese rupees. Consumption is the purchased Food Consumption Score aggregate used throughout the paper; farm production is the matching production aggregate. High-intensity ward is the indicator built from the continuous MMI in the row above it. Difference is the raw gap between the receiving and non-receiving means; the p-value is from a regression of each variable on the remittance indicator. Standard errors are clustered at the district level across 71 districts.

A.1.3. Construction of the household exchange rate

We take the exchange-rate series from Google Finance and cross-validate them against OANDA. For households with migrants in more than one country we average across migrants, following Yang (2008b). That rule binds rarely: among household-years with a foreign migrant, 76.8 percent have exactly one, so the average is that migrant's own rate, and of the 23.2 percent with more than one, a further 56.4 percent send all of them to the same country. Averaging therefore changes the

rate for 10.1 percent of foreign-migrant household-years, 3.3 percent of the estimation sample.⁷ Two coding conventions follow from aggregating to the household. Migrants who moved within Nepal are assigned a rate of one, because their transfers are denominated in rupees and carry no exchange-rate variation, so a household whose migrants are all internal enters at one. A household with no migrant has no destination currency and enters at zero.

A.1.4. Exchange-rate variation

Table A2 reports each destination country with the currency used to construct its rate, the number of migrant person-years in the estimation sample, mean remittances, and the mean, standard deviation and coefficient of variation of the exchange-rate measure over the sample period. Because currencies are quoted in different units, the coefficient of variation is the comparable measure of movement across rows.

India is the single largest destination, accounting for 41 percent of migrants who went abroad, but the peg leaves its rate effectively fixed: its coefficient of variation is 0.00, against 0.02 to 0.19 elsewhere. Indian migrants also remit far less on average, NPR 44,000 against NPR 167,000 to 187,000 for Malaysia and the Gulf states. The identifying variation therefore comes from the non-Indian destinations, and no single one of them dominates it: Malaysia, Qatar, Saudi Arabia and the United Arab Emirates account for 16, 15, 12 and 6 percent of migrants abroad, with coefficients of variation between 0.05 and 0.14, while the remaining high-income destinations add further dispersion. India is reported for completeness, and Bhutan at the Indian rupee rate because the ngultrum is pegged at par. Migrants whose destination is recorded only as “other” cannot be assigned a currency and so fall outside the estimation sample; this costs 615 migrant person-years, 2.9 percent of all migrants in the panel.

7. The comparable figure in Yang (2008b) is 90.2 percent with a single overseas member.

Table A2 – Migrant destinations, remittances, and exchange-rate variation

Destination	Currency	Obs.	Share (%)	Mean remit. (in '000)	Mean ER	SD ER	CV ER
India	INR	3997	41.1	44.4	1.58	0.00	0.00
Malaysia	MYR	1599	16.4	166.9	24.45	1.25	0.05
Qatar	QAR	1486	15.3	181.1	26.41	3.68	0.14
Saudi Arabia	SAR	1194	12.3	184.8	25.92	3.47	0.13
United Arab Emirates	AED	562	5.8	186.7	26.36	3.54	0.13
Japan	JPY	211	2.2	327.0	0.92	0.05	0.05
United States	USD	184	1.9	197.1	84.26	14.97	0.18
United Kingdom	GBP	135	1.4	184.3	118.21	9.56	0.08
Australia	AUD	122	1.3	177.5	70.92	6.98	0.10
South Korea	KRW	113	1.2	542.8	0.09	0.01	0.12
Hong Kong	HKD	79	0.8	92.3	10.89	1.95	0.18
Israel	ILS	24	0.2	338.8	21.76	3.88	0.18
China	CNY	12	0.1	48.9	11.26	2.12	0.19
Bangladesh	BDT	6	0.1	0.0	1.06	0.03	0.02
Bhutan	INR	1	0.0	0.0	1.58	–	–
Total		9725	100.0				

Notes: The unit is the migrant person-year, not the household, so a household with migrants in two countries contributes to two rows. ER is the average annual rate in Nepalese rupees per unit of destination currency before interview, and CV is its coefficient of variation, which makes movements comparable across currencies. Bhutan is shown at the Indian rupee rate because the ngultrum is pegged at par. The Nepalese rupee is itself pegged to the Indian rupee, so India contributes almost no exchange-rate variation despite being the largest destination.

A.1.5. The production distribution

Table A3 reports the distribution of total household production across the household-years in the estimation sample. The concentration of the distribution at low values is what we mean in describing the sample as smallholder: three quarters of household-years produce less than NPR 70,000 of crop and livestock output in a year.

Table A3 – Distribution of total household production

Percentile	25th	50th	75th	90th	95th	99th
Total production (in '000)	7.6	31.1	70.4	132.7	195.8	456.0

Notes: Total household production, crop and livestock, in thousands of 2018 Nepalese rupees.

A.2. Where the identifying variation comes from

The instrument varies across households because their migrants are in different countries and those currencies moved differently against the rupee. Whether that variation, rather than something correlated with it, produces the estimates is what the three exercises below put to the test.

A.2.1. *The reduced form and restricted samples*

Table A4 does two things at once.

Panel B is the benchmark estimate, row by row, on progressively narrower samples. The first row is the full sample and reproduces the main-text estimates: production and the two input margins fall, consumption smoothing and public assistance do not. The rows beneath it drop households by migrant configuration. The exchange rate a household faces depends on where its migrants are, so a single destination could in principle be carrying the result. Excluding households whose migrants are all in India removes the largest destination and moves the production estimate least. Excluding those whose migrants are all internal removes the households that enter the instrument at a rate of one. Excluding both together leaves the smallest sample and the smallest estimate in magnitude, -0.26 against the benchmark -0.42 . Across all four samples farm production stays negative and significant, capital inputs stay negative throughout, and the consumption and public-assistance nulls hold in every one. Labor inputs are the one margin that weakens: significant in the full sample and after excluding India-only migrants, and no longer distinguishable from zero in the two narrowest samples, which drop between a fifth and a third of the household-years. Migrant destination is measured in the same round as the outcome and is itself a household choice, so these samples condition on a post-treatment variable; we read them as diagnostics of where the identifying variation comes from, not as alternative causal estimates.

Panel A relates each outcome directly to the excluded instruments, without dividing by the first stage. It differs from Panel B in two ways that matter for reading it. It involves no first stage, so it cannot be weakened by a weak one; and its coefficients are in units of the exchange rate rather than per log point of remittances, so they are two orders of magnitude smaller and their sizes are not comparable with Panel B or with the main tables. What carries over is the sign pattern: production, capital, and labor all decline, while consumption smoothing and public assistance are

indistinguishable from zero. That last null is the informative one. An exchange rate standing in for some broader local shock should move consumption alongside production, and it does not.

Table A4 — The reduced form and restricted samples

	Dependent variable: $\log(\text{total outcome} + 1)$				
	Farm production (1)	Consumption smoothing (2)	Capital inputs (3)	Labor inputs (4)	Public assistance (5)
<i>Panel A. Reduced form, no first stage</i>					
2016-18 x ER_{hwt} x HI_w	-0.03421*** (0.01022)	0.00163 (0.00389)	-0.03209** (0.01311)	-0.02325** (0.01035)	-0.00415 (0.01143)
<i>Panel B. Triple-difference IV, by estimation sample</i>					
<i>Coefficient on 2016-18 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w</i>					
Full sample (benchmark)	-0.42*** (0.13)	-0.01 (0.05)	-0.41** (0.17)	-0.26** (0.13)	-0.09 (0.11)
Excluding India-only migrants	-0.39*** (0.13)	-0.01 (0.05)	-0.39** (0.16)	-0.23* (0.12)	-0.10 (0.11)
Excluding internal-only migrants	-0.31** (0.14)	0.02 (0.06)	-0.32* (0.17)	-0.18 (0.13)	-0.04 (0.12)
Excluding both	-0.26** (0.13)	0.01 (0.06)	-0.28* (0.14)	-0.12 (0.12)	-0.06 (0.11)
N, full sample	22354	22354	22354	22354	22354
District	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓

Notes: Columns are the five totals on which the paper's argument rests, defined as in Tables 4 to 8. Panel A regresses each outcome directly on the excluded instruments and their interactions, so it involves no first stage; because those coefficients are in units of the exchange rate rather than per log point of remittances, they are reported to five decimal places and their magnitudes are not comparable with Panel B or with the main tables. Panel B reports the instrumented triple difference on four samples. Each drops a set of households: those whose migrants are all in India, those whose migrants are all internal, and both together. Migrant destination is measured in the same round as the outcome and is itself a household decision, so these samples condition on a post-treatment variable and are diagnostics of where the identifying variation comes from rather than alternative causal estimates. The reported N is for the full sample; excluding India-only migrants leaves 19876 household-years, excluding internal-only migrants 18345, and excluding both 15867. All columns are the paper benchmark specification unless the table indicates otherwise: pooled triple difference over 2016–18, $\log(y + 1)$, VDC and year fixed effects, two-stage least squares, with the lower-order remittance terms, the household controls and the ward controls partialled out. Earthquake intensity is measured at the ward. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors, reported in parentheses, are clustered at the district level across 71 districts.

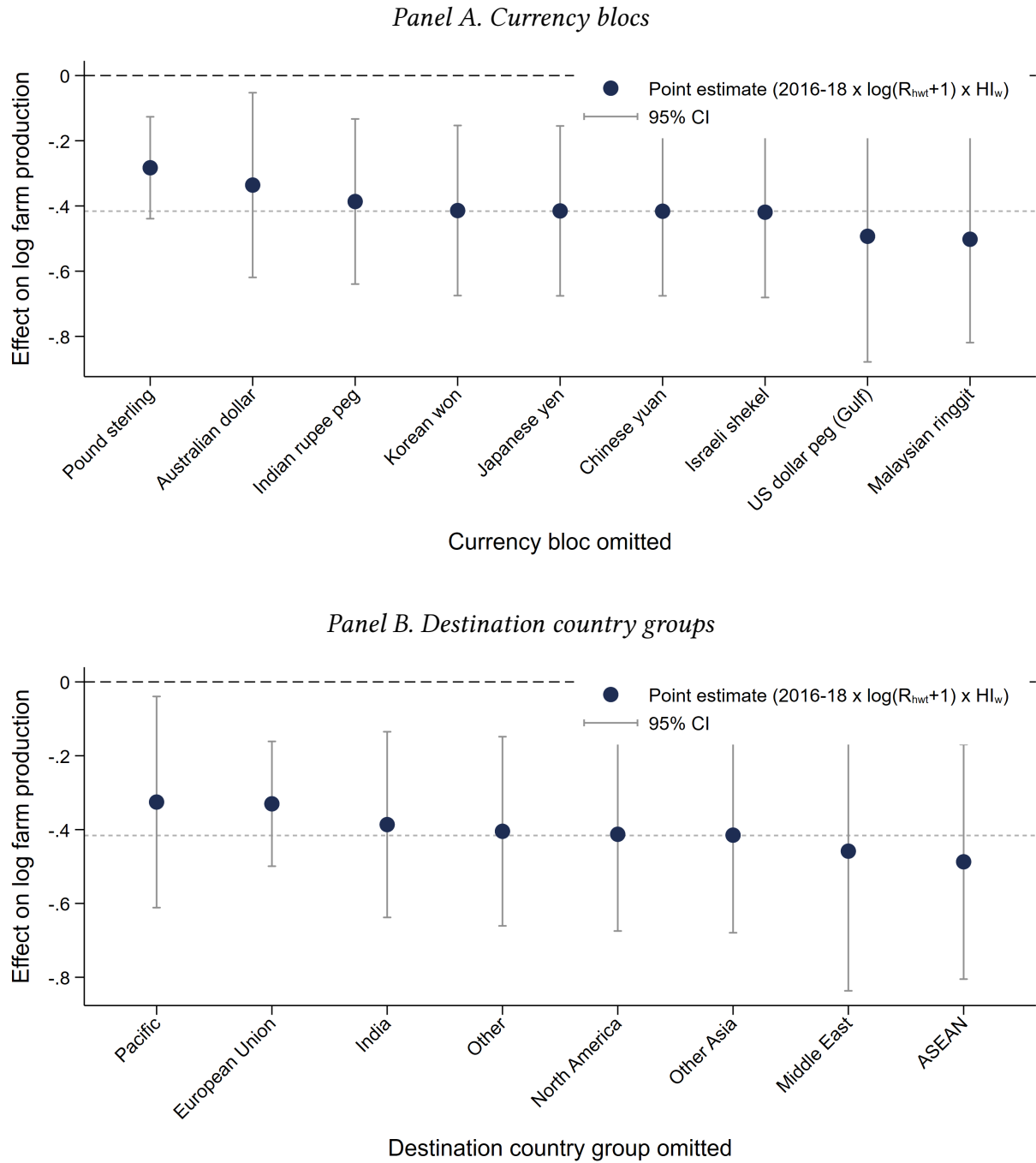
A.2.2. Leaving out one destination

Two further questions concern the destination side of the instrument: whether one destination carries the result, and how the rate is aggregated when a household has migrants in more than one country.

Figure A2 takes the first. Panel A drops one currency bloc at a time and Panel B one destination

country group at a time, re-estimating the benchmark each time. Every estimable group is shown rather than only the large ones. The estimate ranges from -0.28 to -0.50 across blocs and from -0.33 to -0.49 across destination groups, against a benchmark of -0.42 , and no interval in either panel includes zero. Dropping the two largest blocs, the Indian rupee peg and the US dollar peg, moves the estimate least in proportion to the sample it removes. The estimates furthest from the benchmark come instead from dropping very small blocs: omitting sterling, 88 household-years, gives -0.28 , and omitting the Australian dollar, 77 household-years, gives -0.34 . A handful of households therefore carries more weight in the instrument than its share of the sample, which is what showing every bloc makes visible. The two groupings cut the same variation differently and agree.

Figure A2 – Leaving out one destination at a time



Notes: Each point is the triple-difference coefficient $2016-18 \times \log(\widehat{R_{hwt}} + 1) \times HI_w$ from the benchmark specification re-estimated with one group dropped; the bars are 95 percent confidence intervals from district-clustered standard errors. The dashed grey line marks the full-sample benchmark and the dashed black line marks zero. A currency bloc groups migrant destinations that share a currency or a peg. The destination country groups in Panel B are those used to construct the shift-share exposure of [Appendix C](#); a household is assigned the group supplying most of its foreign migrants, with ties broken by the larger total remittance, and households with no foreign migrant are never dropped. The dependent variable is log total farm production.

A.2.3. Alternative rules for aggregating the household exchange rate

The exchange rate is measured at the migrant but the analysis is at the household, so households with migrants in more than one country need an aggregation rule. The paper averages across the household's migrants, the migrant-count weighting of Yang (2008b). That choice binds rarely: 76.8 percent of foreign-migrant household-years have exactly one foreign migrant, and averaging changes the rate for only 3.3 percent of the estimation sample.⁸

Table A5 reports the five totals under the benchmark and under two alternatives that abandon averaging and assign each household the rate faced by a single foreign migrant, the one abroad longest and the oldest. Farm production runs -0.42 , -0.32 and -0.36 , significant at one percent under all three. Capital inputs run -0.41 , -0.34 and -0.38 and labor inputs -0.26 , -0.27 and -0.30 , each significant at five percent throughout. The consumption and public-assistance nulls hold under every rule. No estimate changes sign, and nothing insignificant in the benchmark becomes significant. The two alternatives differ from the benchmark in the same way and for the same reason, so they are not independent tests; what separates them is only the variable used to rank migrants, and on that they agree closely.

8. A further 5.9 percent have one foreign migrant alongside at least one internal migrant. Averaging moves their rate as well, because internal migrants enter at one, and that is precisely what the two alternatives remove.

Table A5 – Alternative rules for aggregating the household exchange rate

	Dependent variable: $\log(\text{total outcome} + 1)$				
	Farm production (1)	Consumption smoothing (2)	Capital inputs (3)	Labor inputs (4)	Public assistance (5)
<i>Coefficient on $2016-18 \times \log(\widehat{R}_{hwt} + 1) \times HI_w$, under each rule</i>					
Average over all migrants (benchmark)	-0.42*** (0.13)	-0.01 (0.05)	-0.41** (0.17)	-0.26** (0.13)	-0.09 (0.11)
Longest-gone foreign migrant	-0.32*** (0.12)	0.01 (0.04)	-0.34** (0.14)	-0.27** (0.14)	-0.17 (0.12)
Oldest foreign migrant	-0.36*** (0.13)	-0.00 (0.05)	-0.38** (0.15)	-0.30** (0.15)	-0.17 (0.12)
N	22354	22354	22354	22354	22354
District	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓

Notes: Columns are the five totals on which the paper’s argument rests, defined as in Tables 4 to 8. Each row keeps the specification fixed and varies only the rule used to assign each household its exchange rate. The first row averages over all of the household’s migrants, which is the migrant-count weighting of Yang (2008b) and is the rule used throughout the paper. The second and third instead assign the household the rate faced by a single foreign migrant, chosen by the longest time spent abroad and by age respectively; in all three rows a household with no foreign migrant carries one if it has an internal migrant and zero otherwise. The averaging rule binds only for the 3.3 percent of household-years whose migrants are in more than one foreign country, since 76.8 percent of household-years with any foreign migrant have exactly one, so the three rules coincide for them. The second and third rows differ from one another only in the variable used to rank migrants, so they are not independent tests. N is reported for the benchmark rule; the estimation sample is the same under all three. The reported coefficient is on $2016-18 \times \log(\widehat{R}_{hwt} + 1) \times HI_w$. The specification, controls, fixed effects and inference are as in Appendix Table A4.

A.3. Inference and Levels of Aggregation

Two features of the design strain conventional inference: the excluded instruments could in principle be weak, and the 71 districts that define the clusters are neither numerous nor evenly treated. The first exercise below addresses both. The two that follow vary the level at which the design operates, the level standard errors are clustered on and the level at which local conditions are absorbed, since each is a choice the benchmark makes rather than one the data force.

A.3.1. Weak-instrument-robust and small-cluster inference

The first stage is strong, but conventional two-stage least squares inference can still be sensitive to identification strength, and district clustering can be sensitive to the number and distribution

of treated clusters rather than the raw count (MacKinnon and Webb 2018). Table A6 addresses both over the same six columns.

Panel A reports Anderson–Rubin tests, valid however weak the instruments are (Anderson and Rubin 1949). The joint null is rejected for production, capital, and labor, and not rejected for consumption smoothing or public assistance, reproducing the pattern of the point estimates without the strong-identification approximation. The two confidence sets differ in what they assume about the nuisance parameters; we treat the joint test as primary and the projection set as the conservative interval. Production excludes zero comfortably at $[-0.873, -0.107]$; capital and the all-items consumption measure exclude it only narrowly, with upper bounds of -0.012 and -0.006 ; labor includes zero.

Panel B reports wild cluster bootstrap p -values alongside the analytic ones. Production and the all-items consumption measure stay strongly significant at 0.002 and 0.003, and total capital at 0.013 with an interval excluding zero. Labor weakens to 0.064 with an interval including zero.

Table A7 applies the same bootstrap to the first stage. The weakest of the four joint F -statistics is 63.5 and all four have bootstrap p -values below 0.001. Its note reports the cluster structure that motivates Panel B: 71 raw district clusters, an effective count of 46.5 once cluster sizes are taken into account, and fewer still among districts containing treated households. Because the endogenous regressor varies continuously within district, identification does not rest on comparisons of treated-cluster means alone, which is why the bootstrap intervals remain informative despite that small effective count.

Table A6 — Weak-instrument-robust and small-cluster inference

	Dependent variable: $\log(\text{total outcome} + 1)$					
	Farm production (1)	Consumption smoothing (2)	Capital inputs (3)	Labor inputs (4)	Public assistance (5)	Consumption all items (6)
2016-18 $\times \log(\widehat{R}_{\text{hwt}} + 1) \times \text{HI}_w$, 2SLS	-0.42*** (0.13)	-0.01 (0.05)	-0.41** (0.17)	-0.26** (0.13)	-0.09 (0.11)	-0.08*** (0.02)
<i>Panel A. Anderson–Rubin, valid under weak instruments</i>						
AR p -value	0.006	0.169	0.047	0.045	0.598	0.035
AR projection set	[-0.873, -0.107]	[-0.114, 0.168]	[-1.011, -0.012]	[-0.630, 0.037]	[-0.365, 0.248]	[-0.122, -0.006]
AR strongly identified set	[-0.741, -0.168]	[-0.097, 0.121]	[-0.839, -0.081]	[-0.545, -0.020]	[-0.308, 0.167]	[-0.113, -0.027]
<i>Panel B. Wild cluster bootstrap, 9999 replications</i>						
Analytic p -value	0.003	0.917	0.020	0.047	0.452	0.000
Bootstrap p -value	0.002	0.923	0.013	0.064	0.480	0.003
Bootstrap 95% CI, lower	-0.741	-0.103	-0.848	-0.556	-0.321	-0.114
Bootstrap 95% CI, upper	-0.153	0.116	-0.068	0.014	0.172	-0.031
N	22354	22354	22354	22354	22354	22354
District	71	71	71	71	71	71

Notes: Columns 1 to 5 are the five totals on which the paper’s argument rests, defined as in Tables 4 to 8. Column 6 adds the wider consumption measure that values home-produced food at market prices. The row above the panels is the two-stage least squares estimate reported elsewhere in the paper, with its district-clustered standard error beneath; both panels test that same coefficient. In Panel A the AR p -value inverts the Anderson–Rubin statistic on all four remittance coefficients jointly (Anderson and Rubin 1949) and is valid however weak the instruments are; it is the primary test. The projection set treats $\log(R + 1)$ and its post interaction as strongly identified, inverts over the remaining two coefficients and projects onto the triple-difference term, and is the conservative interval; the strongly identified set instead assumes all three remaining remittance terms are strongly identified, a stronger assumption that yields a correspondingly narrower set. Both are 95 percent sets. In Panel B the bootstrap p -value and interval come from the wild restricted efficient bootstrap of Davidson and MacKinnon (2010) for two-stage least squares, with Rademacher weights, the null imposed, and resampling at the district level; the analytic p -value above it uses the district-clustered standard error, as in the main tables. First-stage strength under the same bootstrap, and the cluster structure that motivates Panel B, are in Table A7. VDC fixed effects are partialled out and standard errors are clustered on district across 71 districts.

Table A7 – First-stage strength under the bootstrap

	Endogenous regressor			
	$\log(R + 1)$ (1)	$\log(R + 1) \times HI_w$ (2)	$\log(R + 1) \times 2016-18$ (3)	$\log(R + 1) \times HI_w \times 2016-18$ (4)
Joint F , four excluded instruments	123.76	63.53	119.66	67.28
Analytic p -value	0.000	0.000	0.000	0.000
Bootstrap p -value	0.000	0.000	0.000	0.000

Notes: Each column is one of the four endogenous remittance terms, regressed by ordinary least squares on the four excluded exchange-rate instruments together with the controls and VDC and year fixed effects; the reported statistic is the joint F -test of the four instruments. The bootstrap p -value applies the wild cluster bootstrap of Davidson and MacKinnon (2010) at the district level with 9999 replications. The cluster structure that motivates the exercise, following MacKinnon and Webb (2018), is as follows: there are 71 raw district clusters, but the size-adjusted effective number is 46.5, where the effective count is $(\sum_g N_g)^2 / \sum_g N_g^2$; 35 of 71 districts contain treated households and 12 contain both treated and untreated households, and the effective count among treated districts is 25.3. Because the endogenous regressor varies continuously within district, identification does not rest on comparisons of treated-cluster means alone, which is why the bootstrap intervals in Panel B of Table A6 remain informative despite the small effective count among treated districts.

A.3.2. Alternative levels of clustering

Table A8 varies the level of clustering down the rows and reports all five outcomes across the columns. The sample, the specification and the estimates are the benchmark throughout, so the coefficients repeat by construction and only the standard errors and the significance stars move. Clustering on the currency bloc, and two-way on district and bloc, leave the production effect significant at five percent, and leave the pattern across the other four margins unchanged. With only eleven blocs these are bounds on the sensitivity of inference rather than preferred alternatives.

Table A8 – Alternative levels of clustering

	Dependent variable: $\log(\text{total outcome} + 1)$				
	Farm production (1)	Consumption smoothing (2)	Capital inputs (3)	Labor inputs (4)	Public assistance (5)
<i>Coefficient on $2016-18 \times \log(\widehat{R}_{hwt} + 1) \times HI_w$, clustered at each level</i>					
District (benchmark)	-0.42*** (0.13)	-0.01 (0.05)	-0.41** (0.17)	-0.26** (0.13)	-0.09 (0.11)
Currency bloc	-0.42** (0.19)	-0.01 (0.01)	-0.41** (0.18)	-0.26** (0.12)	-0.09 (0.08)
District and bloc (two-way)	-0.42** (0.18)	-0.01 (0.03)	-0.41** (0.20)	-0.26** (0.11)	-0.09 (0.09)
Clusters	71 districts; 11 currency blocs				
N	22354	22354	22354	22354	22354
VDC and year FEs	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓

Notes: Columns are the five totals on which the paper's argument rests, defined as in Tables 4 to 8. Only the clustering of the standard error varies down the rows: the sample, the specification and the estimates are the benchmark throughout, so the coefficients repeat by construction and only the standard errors and the significance stars move. A currency bloc groups migrant destinations sharing a currency or a peg. The bloc count is small by construction, so the second and third rows bound the sensitivity of inference rather than offering a preferred alternative. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The specification, controls and fixed effects are as in Appendix Table A4.

A.3.3. Alternative levels of fixed effects

Section 4.3 sets out why the benchmark absorbs the VDC rather than the ward. The reason is overlap. The rounds are repeated cross-sections drawn from different survey instruments, so the pre- and post-earthquake samples share few localities: only 11 of the 875 wards we observe

appear both before and after the earthquake, against 69 of 712 VDCs, roughly six times larger in household-years. [Figure A3](#) shows what that means geographically. Intensity is measured at the ward, in the top panel, and varies sharply over short distances, while remittances and production in the lower panels are spread across districts. Absorbing the district would discard the within-district contrast that the top panel makes visible; absorbing the ward would discard almost all of the pre-post comparison. The VDC is the finest grouping that retains both.

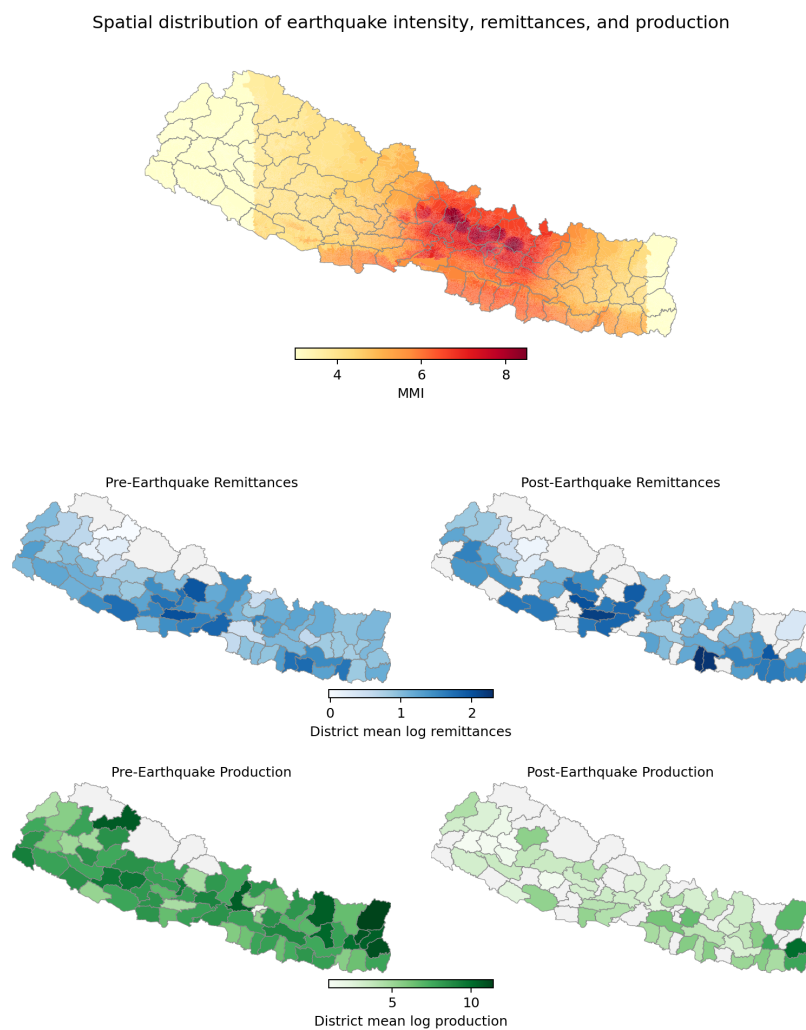
[Table A9](#) goes the coarse way regardless and replaces the VDC fixed effects with district fixed effects. It repeats the final column of each of [Tables 4 to 8](#), so every estimate is directly comparable to its main-text counterpart. The pattern across margins is unchanged: production, capital and labor all fall and remain significant, while consumption and public assistance stay indistinguishable from zero. The estimates are smaller, production at -0.21 , as absorbing the district removes part of the within-district variation in intensity the benchmark uses.

Table A9 – Robustness to district fixed effects

	Dependent variable: $\log(\text{total outcome} + 1)$				
	Farm production (1)	Consumption smoothing (2)	Capital inputs (3)	Labor inputs (4)	Public assistance (5)
2016-18 x HI_w	0.45 (0.39)	-0.16 (0.14)	0.73* (0.41)	-0.19 (0.37)	1.22** (0.57)
2016-18 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.21*** (0.08)	0.01 (0.03)	-0.23** (0.09)	-0.18** (0.08)	-0.02 (0.09)
N	22354	22354	22354	22354	22354
District	71	71	71	71	71
District and year FEs	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓

Notes: Columns are the five totals on which the paper’s argument rests, defined as in [Tables 4 to 8](#). Each column repeats the final column of its main-text counterpart, so the estimates are directly comparable. Two-stage least squares with district fixed effects in place of the VDC fixed effects used elsewhere in the paper. Earthquake intensity is measured at the ward throughout, so it continues to vary within the absorbed unit. The reported coefficient pools the post-earthquake years 2016–18 against the 2011 base. The remittance interactions are instrumented with the matching exchange-rate interactions, which enter in levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors, reported in parentheses, are clustered at the district level across 71 districts.

Figure A3 — Spatial distribution of earthquake intensity, remittances, and production



Notes: The top panel shows ward-level earthquake intensity measured by the Modified Mercalli Intensity (MMI) scale. The lower panels show district-average pre- and post-earthquake remittances and production. Intensity varies within districts, which is what the VDC fixed effects retain and district fixed effects discard. Source: USGS, household-survey data, and authors' calculations.

A.4. Robustness

Each exercise below changes one feature of the benchmark and holds the rest fixed, so the departure from the benchmark is always the one named in the column head or the marker row.

A.4.1. Alternative approaches to dealing with zero outcomes

Zeros make the choice of transformation substantive rather than innocuous (Chen and Roth 2024). Table 3 in the main text reports total production across transformations; Table A10 extends the comparison to every outcome block, under a calibrated log in Panel A and under $\text{lhs}(y)$ in Panel B. The estimates are similar throughout and similar to the benchmark.

Table A10 – Alternative approaches to dealing with zero outcomes

	Dependent variable: transformed total outcome				
	Farm production (1)	Consumption smoothing (2)	Capital inputs (3)	Labor inputs (4)	Public assistance (5)
<i>Panel A. Calibrated-log transformation</i>					
2016-18 x HI_w	0.31 (0.93)	0.06 (0.22)	0.13 (0.82)	-0.82 (0.75)	1.74** (0.77)
2016-18 x $\log(\widehat{R}_{\text{hwt}} + 1) \times \text{HI}_w$	-0.42*** (0.14)	-0.00 (0.05)	-0.42** (0.18)	-0.26** (0.13)	-0.09 (0.11)
<i>Panel B. Inverse hyperbolic sine transformation</i>					
2016-18 x HI_w	0.34 (0.98)	0.05 (0.23)	0.17 (0.87)	-0.85 (0.79)	1.83** (0.81)
2016-18 x $\log(\widehat{R}_{\text{hwt}} + 1) \times \text{HI}_w$	-0.44*** (0.14)	-0.00 (0.05)	-0.45** (0.19)	-0.27** (0.14)	-0.09 (0.12)
N	22354	22354	22354	22354	22354
District	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓

Notes: Columns are the five totals on which the paper's argument rests, defined as in Tables 4 to 8. Both panels re-estimate the benchmark on the same sample and vary only the transformation applied to the outcome. Only the outcome is transformed: the endogenous remittance term remains $\log(1 + R)$ and the excluded instrument the exchange rate in levels, as in the main-text transformation table, so both panels are directly comparable with it and with each other. Calibrated-log specifications assign zero outcomes a value of -0.1, following Chen and Roth. The specification, controls, fixed effects and inference are as in Appendix Table A4.

A.4.2. Alternative measures of treatment

Table A11 varies the definition of earthquake intensity down the rows and reports all five outcomes across the columns. The first row is the benchmark, which calls a ward high-intensity when its MMI exceeds the sample median of 5.2; because MMI is recorded in increments of 0.2, the operative threshold is 5.4.

Farm production is negative and significant under every definition. Standardized continuous intensity gives -0.19 per standard deviation of shaking, raising the cutoff to 6.0 gives -0.27 and to 6.4, the most heavily shaken fifth of the sample, gives -0.36 . A donut specification dropping the two bins adjacent to the threshold gives -0.49 against the benchmark -0.42 , so the result is not manufactured by wards near the cut. Consumption smoothing is indistinguishable from zero in all five rows, which is the null the main text relies on and the one least sensitive to how treatment is defined.

The input margins are negative in every row and significant under the benchmark, the continuous measure and the donut, at -0.41 , -0.16 and -0.47 for capital and -0.26 , -0.11 and -0.37 for labor. They lose significance at the two higher cutoffs, which retain 8,258 and 4,801 treated household-years against 10,881: narrowing the treated group costs precision, and we report that rather than the definitions that happen to survive. The median split is the definition the design is built on, and these rows vary it to show how far the estimates travel, not to choose among them. We keep the binary indicator at the median because it is transparent, follows Pathak and Schündeln (2026), and retains wards with no intensity data on the control side of the cut.

Table A11 – Alternative measures of treatment

	Dependent variable: $\log(\text{total outcome} + 1)$				
	Farm production (1)	Consumption smoothing (2)	Capital inputs (3)	Labor inputs (4)	Public assistance (5)
<i>Coefficient on $2016-18 \times \log(\widehat{R}_{\text{hwt}} + 1) \times \text{HI}_w$, under each intensity definition</i>					
MMI ≥ 5.4 (benchmark)	-0.42*** (0.13)	-0.01 (0.05)	-0.41** (0.17)	-0.26** (0.13)	-0.09 (0.11)
Standardized continuous MMI	-0.19** (0.08)	-0.01 (0.02)	-0.16** (0.08)	-0.11* (0.06)	-0.04 (0.05)
MMI ≥ 6.0 (higher cutoff)	-0.27** (0.13)	-0.02 (0.04)	-0.10 (0.15)	-0.15 (0.10)	-0.15 (0.10)
MMI ≥ 6.4 (top fifth)	-0.36** (0.14)	-0.02 (0.04)	-0.14 (0.14)	-0.21** (0.10)	-0.19* (0.11)
Donut, dropping MMI 5.2 and 5.4	-0.49*** (0.13)	0.03 (0.05)	-0.47** (0.18)	-0.37*** (0.13)	-0.05 (0.12)
District	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓

Notes: Columns are the five totals on which the paper's argument rests, defined as in Tables 4 to 8. Each row redefines earthquake intensity and rebuilds every interaction and excluded instrument from that definition, so each is a complete re-estimation of the benchmark rather than a reweighting of it. Only the triple-difference coefficient is reported: the lower-order difference-in-differences term is not comparable across rows, because in the second row it is a slope per standard deviation of shaking rather than a contrast between groups. MMI is recorded in increments of 0.2, so although the sample median is 5.2 the treated group begins at 5.4, and the benchmark cutoff is stated at that effective value. The final row keeps the benchmark indicator but drops the two intensity bins straddling the threshold, 5.2 below it and 5.4 above it, which tests whether the result is manufactured by wards adjacent to the cut. Every row uses 22354 household-years except the last, where the donut restriction leaves 20406. Treated household-years are 10881, 8258, 4801 and 10016 in rows one, three, four and five; the second row has no treated group because intensity there is continuous. The specification, controls, fixed effects and inference are as in Appendix Table A4.

A.4.3. Alternative measures of farm production

Table A12 decomposes the production effect into a quantity component and a price component across columns. Column 1 is the benchmark, production at local prices. Columns 2 and 3 value production at fixed national prices, identical in every year and every ward, so no price movement can enter; column 4 measures output in physical kilograms and removes prices altogether. The estimate is -0.37 and -0.42 under the two fixed-price measures and -0.24 in kilograms, each significant at one percent; quantities are in different units from values, so what carries across column 4 is the sign and significance rather than the magnitude. Column 5 isolates the residual

price component, the log ratio of value at local prices to value at fixed prices, which is precisely estimated at zero. Because $\log(\text{value at local prices}) = \log(\text{value at fixed prices}) + \log(\text{price factor})$ household by household, these columns decompose the published effect exactly rather than approximating it.

Most crop rows carry no price reported by the household itself. The share with an own price rises from 10 percent in 2011 to 23 percent in 2018 and is 17 percent overall, with the rest valued by an imputation cascade running from the household's own price through the primary sampling unit, the VDC and the district to the national median. An imputed local price is still a local price, which is why the fixed-price exercise is the decisive test rather than a supplementary one.

Table A12 — Is the production effect quantity or price?

	Dependent variable:				
	Total production local prices (benchmark) (1)	Crop production fixed national prices (2)	Crop at fixed prices plus livestock (3)	Physical harvest in kilograms (4)	Price factor local over fixed (5)
2016-18 x HI_w	0.30 (0.92)	0.18 (0.89)	0.22 (0.88)	0.01 (0.63)	-0.00 (0.05)
2016-18 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.42*** (0.13)	-0.37*** (0.13)	-0.42*** (0.14)	-0.24*** (0.09)	0.00 (0.01)
N	22354	22354	22354	22354	22354
District	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓

Notes: All columns are in logs of one plus the stated quantity. Fixed national prices value every crop-unit pair at the pooled sample median unit value, identical in every year and every ward, so no price movement of any kind can enter. Because $\log(\text{value at local prices}) = \log(\text{value at fixed prices}) + \log(\text{price factor})$ household by household, the columns decompose the published effect exactly: column 1 is the benchmark, columns 2 to 4 hold prices fixed, and column 5 isolates the residual price component. Whether the instrument moves a local price index at all is tested separately in Table A13. The specification, controls, fixed effects and inference are as in Appendix Table A4.

The companion question is whether the instrument moves a local price at all. Table A13 answers it. Each column is a fixed-basket price index built at a different level, so the column head names the level at which the index is computed, not a different outcome concept: a VDC-by-year index, a district-by-year index, and the household's own selling price. A fixed basket means the index moves only when the prices facing a household move, and not when the household changes what it produces, which is what separates this table from the one above. Panel A instruments the remittance interactions as in the rest of the paper; Panel B regresses the index directly on the excluded instruments and so involves no first stage. Neither the VDC-year nor the district-year

index responds under either estimator, and the household's own selling price does not either. The instrumented VDC-year coefficient is -0.0020 with a standard error of 0.0023 ; it would have to be roughly two hundred times its estimated magnitude to account for the -0.42 production effect. Together the two tables say that the production response is a quantity response.

Table A13 – Does the instrument move local prices?

	Dependent variable: log fixed-basket price index		
	VDC by year (1)	District by year (2)	Household's own selling price (3)
<i>Panel A. Triple-difference IV</i>			
2016-18 x $\log(\widehat{R}_{\text{hwt}} + 1)$ x HI_w	-0.0020 (0.0023)	-0.0029 (0.0022)	-0.0139 (0.0239)
<i>Panel B. Reduced form, no first stage</i>			
2016-18 x ER_{hwt} x HI_w	-0.00015 (0.00026)	-0.00020 (0.00020)	-0.00207 (0.00271)
N	13344	20872	6734
District	71	71	71
VDC and year FEs	✓	✓	✓
Ψ_{hwt}	✓	✓	✓

Notes: Each column is a fixed-basket price index built at the stated level, so it moves only when the prices facing a household move, not when the household changes what it produces. The first row instruments the remittance interactions as in the rest of the paper; it is reported to four decimal places rather than the two used elsewhere because the estimates are two orders of magnitude smaller than the production effect. The second row regresses the index directly on the excluded instruments, which involves no first stage; because those coefficients are in units of the exchange rate rather than log remittances, they are reported to five decimal places and are not comparable in magnitude with the first row. Sample sizes differ across columns because a VDC-year index, a district-year index and a seller's own price are available for different sets of households. The specification, controls, fixed effects and inference are as in Appendix Table A4.

A.4.4. Excluding the Kathmandu Valley

Table A14 excludes Kathmandu, Lalitpur and Bhaktapur. The valley is the least agricultural part of the sample, was heavily shaken, and holds Nepal's largest urban concentration. The pattern across margins is unchanged: production, capital and labor all fall and remain significant, and consumption and public assistance remain indistinguishable from zero. The production estimate is smaller, -0.22 against the benchmark -0.42 , so the valley contributes to the magnitude without driving the result.

Table A14 – Robustness to excluding Kathmandu Valley districts

	Dependent variable: $\log(\text{total outcome} + 1)$				
	Farm production (1)	Consumption smoothing (2)	Capital inputs (3)	Labor inputs (4)	Public assistance (5)
2016-18 x HI_w	-0.14 (0.82)	0.01 (0.21)	-0.26 (0.74)	-1.02 (0.63)	1.52* (0.77)
2016-18 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.22*** (0.08)	0.02 (0.04)	-0.25*** (0.09)	-0.18** (0.08)	0.02 (0.09)
N	21383	21383	21383	21383	21383
District	68	68	68	68	68
VDC and year FEs	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓

Notes: Columns are the five totals on which the paper’s argument rests, defined as in Tables 4 to 8. Kathmandu, Lalitpur and Bhaktapur are excluded: the valley is the least agricultural part of the sample, was heavily shaken, and holds Nepal’s largest urban concentration. The specification, controls and fixed effects are as in Appendix Table A4. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors, reported in parentheses, are clustered at the district level across the 68 districts that remain once the Kathmandu Valley is excluded.

A.5. Further results

The three exercises below fill in margins the main text summarizes without displaying in full: consumption measured over all household items, the composition of the labor response, and the year-by-year estimates behind the pooled interaction.

A.5.1. Total consumption

The main text measures consumption by what the household purchases, because home-produced food consumed is by construction a line item inside the production outcome and production and consumption decisions are non-separable in this setting (de Janvry, Fafchamps, and Sadoulet 1991). This subsection widens the measure.

Table A15 reports the same specification over all household consumption items, including those produced at home and valued at market rates. On this wider measure the response is negative, at 0.08. The magnitude is what matters: it is roughly a fifth of the 0.42 production response, so even the measure that shares rupees with farm production shows the production margin adjusting several times more than the consumption margin.

Table A16 repeats the mediation decomposition of Appendix B with consumption measured

Table A15 – Consumption components and total, all household consumption items

	Dependent variable: log(consumption variables, all items + 1)							
	Grains (1)	Pulses (2)	Vegetables (3)	Fruit (4)	Meat (5)	Milk and eggs (6)	Oil (7)	Total (8)
2016-18 x HI_w	0.12 (0.12)	0.26 (0.32)	0.05 (0.16)	1.38** (0.58)	0.26 (0.57)	-0.92 (0.85)	0.07 (0.17)	0.06 (0.10)
2016-18 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.07*** (0.02)	-0.14 (0.09)	-0.06* (0.03)	-0.26* (0.15)	0.08 (0.17)	-0.20 (0.17)	-0.05 (0.03)	-0.08*** (0.02)
N	22354	22354	22354	22354	22354	22354	22354	22354
District	71	71	71	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Columns are the seven FCS groups and their total, measured over all household consumption items rather than purchased items alone, so home-produced food valued at market prices is included. That measure shares rupees with the production outcome by construction, which is why the main text uses the purchased measure in Table 5. The specification, controls, fixed effects and inference are as in Appendix Table A4.

the same wider way. The consumption share rises, but the increase is an accounting artifact rather than a behavioral one, for the reason just given: an indirect effect running through home-produced consumption shares rupees with the outcome. We report it for transparency and interpret the main-text column.

Table A16 — IV mediation of total production through transformed totals, all consumption items

	Dependent variable: log(total production + 1)				
	Farm production	Consumption smoothing	Capital inputs	Labor inputs	Public assistance
Total effect (TD)	-0.42*** (0.14)	-0.42*** (0.14)	-0.42*** (0.14)	-0.42*** (0.14)	-0.42*** (0.14)
Indirect effect		-0.08*** (0.02)	-0.30*** (0.16)	-0.18*** (0.11)	-0.00 (0.00)
Direct effect		-0.34*** (0.14)	-0.12 (0.08)	-0.23*** (0.08)	-0.42*** (0.14)
Proportion mediated		19.36*** (7.32)	71.05*** (22.49)	44.39*** (17.54)	0.03 (0.22)
N	22354	22354	22354	22354	22354

Notes: The outcome is log(total farm production + 1), summed over the seven FCS food groups the survey records on the production side (WFP 2024). Every column is a household total and every column but consumption smoothing repeats Table 9 exactly; only the consumption mediator changes. Here consumption smoothing is measured over all household consumption items, including those produced at home, whereas the main-text table uses the food the household buys. Items produced and consumed at home are 33 to 40 percent of consumption value and are by construction part of the production outcome, so the indirect effect through this mediator is partly an accounting identity rather than a behavioral channel. The main-text table decomposes the same channel through the bought measure, which carries no such overlap. Standard errors are bootstrap standard deviations across 50 district-clustered replications, and stars follow two-sided sign-based bootstrap p-values. Proportion mediated is in percent and is unstable when the total effect is near zero in bootstrap draws. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The bootstrap resamples across 71 districts.

A.5.2. Labor allocation

Table A17 splits the labor margin of Section 5 into wage and own-account work inside and outside agriculture. The decline is on own agricultural labor; wage work, in or out of agriculture, does not respond. That is the composition behind the labor-input total, and it is consistent with the year-by-year labor estimates reported below.

Table A17 – Labor allocation across agriculture and non-agriculture

	Dependent variable: log(labor-allocation variables + 1)				
	Wage agricultural (1)	Wage non-agricultural (2)	Own agricultural (3)	Own non-agricultural (4)	Total (5)
2016-18 x HI_w	0.31 (0.34)	0.66 (0.53)	-0.62 (0.42)	0.76* (0.45)	-0.33 (0.33)
2016-18 x $\log(\widehat{R_{hwt}} + 1)$ x HI_w	-0.17*** (0.05)	-0.14 (0.16)	-0.15* (0.08)	0.04 (0.16)	0.13 (0.15)
N	22354	22354	22354	22354	22354
District	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓

Notes: Columns split the labor margin into wage and own-account work inside and outside agriculture, each in logs of one plus the value. The specification, controls, fixed effects and inference are as in Appendix Table A4.

A.5.3. Individual post-earthquake years

The mechanism tables report one interaction pooling 2016, 2017 and 2018. Tables A18 to A21 enter each post-earthquake year separately against the 2011 base. The capital timing reported in Section 5 holds component by component: livestock inputs and handling and upkeep follow the same profile as the total, concentrated in the first two rounds and indistinguishable from zero by 2018. Total labor input is negative in every year and significant in 2017 at -0.34 , while hired labor is positive and insignificant throughout, at 0.08 , 0.12 and 0.09 . Consumption shows no year pattern: one of its twenty-four coefficients reaches five percent. Earthquake aid is -0.08 in 2016, significant at ten percent, and 0.01 and 0.02 in the two following years. The pooled estimates therefore average a response that is present in each year and largest in the two rounds immediately after the shock, not one confined to a single round.

Table A18 – Consumption components by post-earthquake year

	Dependent variable: log(consumption variables + 1)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Grains	Pulses	Vegetables	Fruit	Meat	Milk and eggs	Oil	Total
2016 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	0.10 (0.18)	-0.24 (0.18)	0.01 (0.11)	-0.07 (0.22)	0.12 (0.18)	-0.10 (0.20)	-0.06 (0.10)	-0.01 (0.05)
2017 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	0.01 (0.17)	-0.12 (0.19)	0.26** (0.11)	-0.14 (0.15)	0.09 (0.20)	0.06 (0.18)	-0.11 (0.10)	0.01 (0.05)
2018 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.02 (0.19)	-0.17 (0.17)	0.05 (0.08)	-0.20 (0.20)	0.29 (0.20)	-0.18 (0.18)	-0.05 (0.11)	-0.00 (0.05)
N	22354	22354	22354	22354	22354	22354	22354	22354
District	71	71	71	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Consumption is the purchased-food measure of Table 5. The specification, controls, fixed effects and inference are as in Table 4. Each row reports a separate post-earthquake year interaction, with 2011 the omitted base year; the main text reports the pooled version.

Table A19 – Capital input components by post-earthquake year

	Dependent variable: log(capital input variables + 1)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Seed	Fertilizer and pesticide	Irrigation	Equipment rental	Handling and upkeep	Livestock	Total
2016 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.16 (0.12)	-0.19* (0.11)	-0.03 (0.08)	-0.09 (0.14)	-0.28*** (0.10)	-0.35** (0.16)	-0.51*** (0.18)
2017 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.16 (0.11)	-0.14 (0.09)	-0.08 (0.11)	-0.19 (0.15)	-0.21** (0.10)	-0.44** (0.18)	-0.45** (0.18)
2018 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.24* (0.13)	0.01 (0.15)	-0.06 (0.11)	-0.04 (0.14)	-0.10 (0.15)	-0.30 (0.18)	-0.27 (0.20)
N	22354	22354	22354	22354	22354	22354	22354
District	71	71	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓	✓	✓

Notes: Hired labor is reported with labor inputs and is excluded from the capital total. The specification, controls, fixed effects and inference are as in Table 4. Each row reports a separate post-earthquake year interaction, with 2011 the omitted base year; the main text reports the pooled version.

Table A20 – Labor input components by post-earthquake year

	Dependent variable: log(labor input variables + 1)		
	(1)	(2)	(3)
	Hired labor	Own agricultural labor	Total
2016 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	0.08 (0.11)	-0.22 (0.14)	-0.22 (0.14)
2017 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	0.12 (0.14)	-0.33** (0.14)	-0.34** (0.14)
2018 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	0.09 (0.13)	-0.20 (0.13)	-0.22 (0.13)
N	22354	22354	22354
District	71	71	71
VDC and year FEs	✓	✓	✓
Ψ_{hwt}	✓	✓	✓

Notes: The specification, controls, fixed effects and inference are as in Table 4. Each row reports a separate post-earthquake year interaction, with 2011 the omitted base year; the main text reports the pooled version.

Table A21 – Public assistance components by post-earthquake year

	Dependent variable: log(public assistance + 1)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Earthquake aid	Old age and widow	Disability and ethnic	Maternal and child	Conflict and martyr	Other aid	Total
2016 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	-0.08* (0.04)	-0.08 (0.11)	0.00 (0.02)	0.07* (0.04)	0.01* (0.00)	0.00 (0.00)	-0.11 (0.12)
2017 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	0.01 (0.05)	-0.19* (0.11)	-0.02 (0.02)	0.02 (0.03)	0.00 (0.00)	-0.01 (0.01)	-0.15 (0.13)
2018 x $\log(\widehat{R}_{hwt} + 1)$ x HI_w	0.02 (0.05)	-0.19 (0.13)	0.02 (0.02)	0.06 (0.04)	0.00 (0.00)	0.01 (0.01)	-0.08 (0.13)
N	22354	22354	22354	22354	22354	22354	22354
District	71	71	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓	✓	✓
Ψ_{hwt}	✓	✓	✓	✓	✓	✓	✓

Notes: The specification, controls, fixed effects and inference are as in Table 4. Each row reports a separate post-earthquake year interaction, with 2011 the omitted base year; the main text reports the pooled version.

B — The Mediation Decomposition

This appendix section formalizes the mediation exercise reported in Table 9. The setup follows the decomposition logic in Imai, Keele, and Tingley (2010), but is adapted to our triple-difference IV specification and should therefore be read as a structured IV mediation analysis rather than a fully nonparametric mediation design.

Let Y_{hwt} denote transformed total farm production, and let M_{hwt}^k denote one candidate mediator, where $k \in \{\text{consumption total, capital total, labor total, public-assistance total}\}$. For each mediator, we estimate three equations. First, the total effect equation is

$$Y_{hwt} = \alpha_{v(w)} + \tau_t + \beta_1 (Post_t \times HI_w \times R_{hwt}) + \Psi_{hwt} + \varepsilon_{1,hwt}, \quad (\text{B.1})$$

where Ψ_{hwt} is as in (2), collecting the lower-order remittance terms together with the household and ward controls.

Second, for each mediator M_{hwt}^k we estimate

$$M_{hwt}^k = \alpha_{v(w)} + \tau_t + \beta_{2k} (Post_t \times HI_w \times R_{hwt}) + \Psi_{hwt} + \varepsilon_{2k,hwt}. \quad (\text{B.2})$$

Third, we include the mediator directly in the production equation:

$$Y_{hwt} = \alpha_{v(w)} + \tau_t + \beta_{3k} (Post_t \times HI_w \times R_{hwt}) + \xi_k M_{hwt}^k + \Psi_{hwt} + \varepsilon_{3k,hwt}. \quad (\text{B.3})$$

In all three equations, the remittance interaction terms are instrumented with the corresponding exchange-rate interaction terms, exactly as in the main IV specification. For mediator k , the reported decomposition is

$$TE_k = \beta_1, \quad NIE_k = \beta_{2k} \xi_k, \quad NDE_k = \beta_{3k}, \quad PM_k = 100 \times \frac{NIE_k}{TE_k}.$$

Standard errors are obtained by district-clustered bootstrap.

A causal mediation interpretation would require a sequential-ignorability condition for the mediator-outcome relationship (Imai, Keele, and Tingley 2010): conditional on treatment, controls,

and fixed effects, no unobserved factor may jointly affect the mediator and potential production. That is demanding here, because the mediators in (B.3) are post-earthquake responses rather than predetermined covariates, and recovery needs, labor-market disruption, or support from outside the formal remittance channel could move both. We therefore read Table 9 as showing which margins are most strongly associated with the production effect within a disciplined IV decomposition, not as a causal channel decomposition, and alongside the reduced-form evidence in the main text.

C – Predetermined Migration Exposure and Land Cover

The shift-share analysis measures a ward’s predetermined exposure to migrant networks whose destination composition was already changing before the earthquake. For each ward w , let $z_{wc,2001}$ denote its baseline household exposure to destination group c , measured as the share of the ward’s households that reported a migrant in c at the 2001 census (Central Bureau of Statistics 2001); these shares are incomplete, in that they sum to the ward’s overall 2001 migration rate of about 0.14 rather than to one, which is why every specification also controls for that share sum interacted with year. The destination groups are India, other SAARC countries, ASEAN, the Middle East, other Asia, the European Union, other Europe, North America, the Pacific, and other destinations including Africa and the Americas. The exposure is larger when a ward had stronger migrant links to destination groups that subsequently grew more rapidly elsewhere in Nepal. Formally,

$$SS_w = \sum_c z_{wc,2001} G_{c,-d,2001-11},$$

where $G_{c,-d,2001-11}$ is destination-specific migration growth between the two pre-earthquake census waves, calculated outside ward w ’s district d . Nepal had 75 districts during this period, so each ward’s shifter is estimated from destination-specific migration growth in the other 74 districts. This leave-one-district-out growth factor prevents a ward’s own local migration trajectory from mechanically determining its predicted exposure.

This construction follows the exposure logic of Khanna et al. (2026). The interpretation of a Bartik design depends on whether credibility is assigned to predetermined local shares (Goldsmith-

Pinkham, Sorkin, and Swift 2020) or to quasi-random shocks (Borusyak, Hull, and Jaravel 2022). As emphasized in the practical shift-share guidance of Borusyak, Hull, and Jaravel (2025), these designs require a clear account of which variation is plausibly exogenous. In this setting, the design is most naturally interpreted as predetermined exposure to changing migrant networks rather than as a many-shock design. District-clustered standard errors also do not address arbitrary residual correlation across wards with similar exposure shares (Adão, Kolesár, and Morales 2019). We therefore treat the land-cover results as supporting evidence on the timing and production margin, not as a stand-alone migration IV.

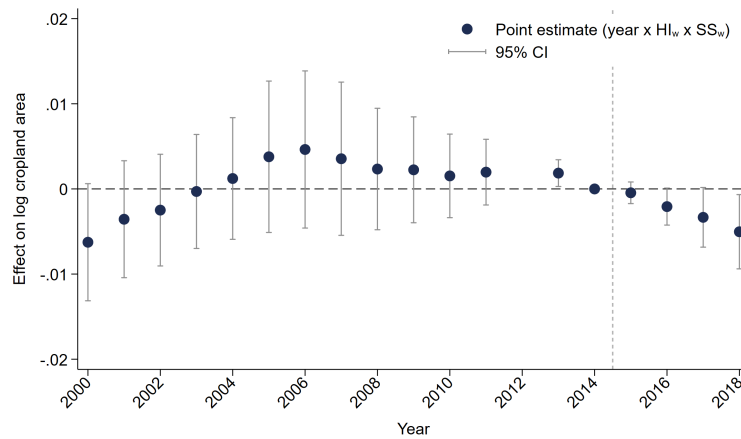
C.1. Continuous exposure at the ward

The main-text specification uses indicator variables for high earthquake intensity and high shift-share migration exposure. Figure C1 and Table C1 show that the pattern is not driven by the binary shift-share cutoff. Instead, they retain the high-intensity earthquake indicator and enter shift-share exposure continuously. The continuous shift-share measure captures differences across wards in the strength of their pre-earthquake migrant links to destination groups whose migration networks were expanding elsewhere in Nepal, analogous to the origin-by-destination exposure logic in Khanna et al. (2026) and the destination-shock remittance logic in Yang (2008b). Each measure is standardized over the 2014 ward cross-section, so the triple interaction represents the differential effect of a one-standard-deviation increase in predicted migration exposure among high-earthquake-intensity wards.

The continuous shift-share specification supports the same qualitative pattern. High-earthquake-intensity wards with stronger predicted migration exposure experience larger post-earthquake declines in cropland. The year-specific estimates become increasingly negative from 2015 through 2018. This pattern is similar across the logarithmic, inverse-hyperbolic-sine, and calibrated-log transformations, while the cropland level estimates are less precise.

The result is also similar when the exposure is expressed as predicted growth rather than the predicted level, which indicates that the pattern is not driven by baseline migration intensity alone.

Figure C1 — Cropland event study with high earthquake intensity and continuous migration shift-share exposure



Notes: Year-specific coefficients on $\text{year} \times HI_w \times SS_w$, where HI_w indicates above-median ward MMI and SS_w is continuous migration shift-share exposure, standardized over the 2014 ward cross-section. The omitted year is 2014 and the dependent variable is $\log(C + 0.09)$, where C is ward cropland hectares. Table C1 gives the full specification.

Table C1 — High-intensity and continuous migration shift-share exposure robustness

	Dependent variable: alternative transformations of cropland area C						
	$\log(C + 0.09)$	$\text{lhs}(C)$	C (ha)	$\log(C)$	calibrated $\log(C)$		
				for $C > 0$	$x=0.1$	$x=0.5$	$x=1$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
2015–18 x HI_w	0.004 (0.006)	0.003 (0.006)	0.982 (0.683)	0.004 (0.006)	0.003 (0.006)	0.003 (0.006)	0.003 (0.006)
2015–18 x $HI_w \times SS_w$	-0.003** (0.001)	-0.003*** (0.001)	-0.174** (0.084)	-0.004*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003** (0.001)
N	617922	617922	617922	611598	617922	617922	617922
District	75	75	75	75	75	75	75
Ward and district-year FE	✓	✓	✓	✓	✓	✓	✓
Core controls	✓	✓	✓	✓	✓	✓	✓

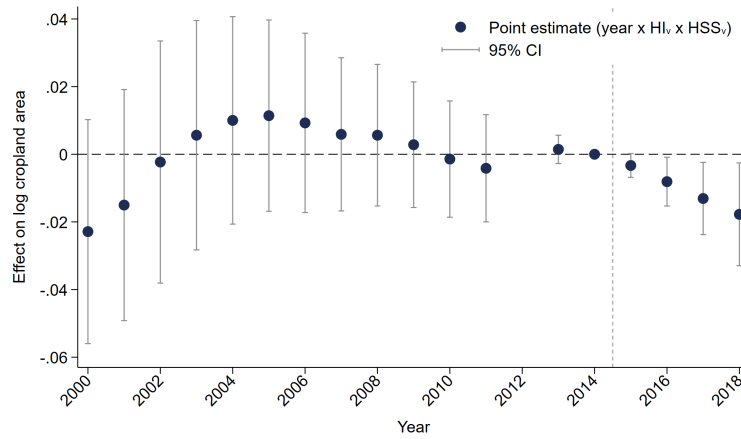
Notes: HI_w indicates above-median ward MMI and SS_w is standardized continuous migration shift-share exposure, built from pre-earthquake destination shares and leave-one-district-out destination growth. C is ward cropland hectares. The sample runs 2001–18 and omits 2014; the coefficient pools the 2015–18 triple interaction and the year-specific estimates are in Figure C1. All columns include ward and district-year fixed effects, year-interacted core controls, and year interactions with baseline destination shares. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors, in parentheses, are clustered on district across 75 districts.

C.2. Aggregating the land-cover design to the VDC

The land-cover evidence is measured at the ward, while the household design absorbs the VDC and controls for ward characteristics, because too few wards appear in both the pre- and post-earthquake surveys to support a ward effect. Running the land-cover exercise at the VDC as well puts both parts of the paper on the same geography and shows that the cropland pattern is not an artifact of the unit at which local conditions are absorbed. It does not test the household design directly, which rests on different data and a different source of variation, but it removes geography as an explanation for the two parts pointing the same way. Cropland and the count-like covariates are summed across the wards of a VDC; intensity, the poverty index and the agricultural household share are averaged. Both group indicators are re-cut at the VDC median in the base year rather than inherited from the ward split, so group membership is redefined at this level rather than carried over from a finer one.

Figure C2 is the VDC counterpart of Figure C1 and plots the year-specific coefficients. The path is the one the ward-level figure shows. Before the earthquake the estimates are imprecise and indistinguishable from zero throughout, with no trend running into 2014. From 2015 they turn steadily more negative, reaching about -0.018 by 2018, and the intervals exclude zero in 2017 and 2018. Table C2 reports the same specification across transformations; the pooled triple interaction remains negative and significant at five percent throughout. Aggregating the design to the geography the household regressions absorb therefore does not weaken it.

Figure C2 – Cropland event study at the VDC level



Notes: Year-specific coefficients on $\text{year} \times HI_v \times HSS_v$, with 95 percent confidence intervals; the vertical bar marks the 2015 earthquake. Both indicators are cut at the VDC median in the base year rather than inherited from the ward split. The omitted year is 2014 and the dependent variable is $\log(C + 0.09)$, where C is VDC cropland hectares (NLCMS 2022). This is the VDC counterpart of Figure C1; Table C2 gives the full specification.

Table C2 – Cropland and migration shift-share exposure at the VDC level

	Dependent variable: alternative transformations of cropland area C						
	$\log(C + 0.09)$	$\text{lhs}(C)$	C (ha)	$\log(C)$	calibrated $\log(C)$		
				for $C > 0$	$x=0.1$	$x=0.5$	$x=1$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
2015–18 x HI_v	0.011* (0.006)	0.011* (0.006)	12.303* (6.341)	0.012* (0.006)	0.011* (0.006)	0.011* (0.006)	0.011* (0.006)
2015–18 x $HI_v \times HSS_v$	-0.011** (0.004)	-0.011** (0.004)	-6.036* (3.318)	-0.010** (0.004)	-0.011** (0.004)	-0.011** (0.004)	-0.011** (0.004)
N	69822	69822	69822	69665	69822	69822	69822
District	75	75	75	75	75	75	75
VDC and district-year FE	✓	✓	✓	✓	✓	✓	✓
Core controls	✓	✓	✓	✓	✓	✓	✓

Notes: The land-cover exercise of Section 4.2 repeated at the VDC. Cropland and the count-like covariates are summed across the wards of a VDC; intensity, poverty and the agricultural share are averaged. HI_v and HSS_v are cut at the VDC median in the base year. The coefficient pools 2015–18; year-specific estimates are in Figure C2. All columns include VDC and district-year fixed effects, year-interacted core controls, and year interactions with baseline destination shares. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors, in parentheses, are clustered on district across 75 districts.